

RESEARCH ARTICLE

Cross-shelf exchange in the shelf of the East China Sea

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Key Points:

- ECS boundary exchanges and further delivery of mixture water are quantified
- Places with persistent and dynamic exchange are located along the shelf break
- Mechanisms driving the exchanges are analyzed

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Abstract A high-resolution, 3-dimensional primitive equation model is used to investigate the cross-shelf exchange in the East China Sea (ECS). Favorable comparisons between field data and model simulations from both climatological run and hindcast run for 2006 indicate that the model has essential skills in capturing the key physics of the ECS. Temporal and spatial variations of the cross-shelf exchanges are further analyzed. It was demonstrated from both observations and simulations that in 2006 high saline water could be delivered to the north of the Changjiang River mouth (near 32°N) as a result of stronger than typical cross-shelf exchanges at the shelf break and flows through the Taiwan Strait with an annual mean rate of 2.59 and 1.83 Sv, respectively. A few new places at the shelf break were also identified where persistent and vigorous onshore or offshore exchanges occur throughout the year. Cross-shelf exchange is largely determined by the along-shelf geostrophic balance with weak seasonality, which is modulated in upper layers by northeasterly monsoon from early-fall to late-spring and at seabed by bottom friction during December–January, May, and August–September. Nonlinear effect, with strong spatial variations and intra-seasonal variability, is a secondary but persistent contributor to the net seaward transport, except for north-east of Taiwan where the nonlinear effect becomes significant but more varied.

1. Introduction

The shelf of the East China Sea (ECS) sustains an enhanced primary productivity due to rich supply of nutrients from both terrestrial and marine sources. The Changjiang (Yangtze) River (CR) and several other rivers are major contributors of terrestrial nutrients [Tian *et al.*, 1993; Zhang, 1996]. Among them, the CR supplies about 80% of the total discharge of fresh water in which nitrate and silicate concentration reaches as high as 70 and 100 $\mu\text{mol L}^{-1}$, respectively, with significant seasonality [Edmond *et al.*, 1985]. For the latest decade, the growing trend of nitrate is rather obvious while the silicate appears to be slowly declining due to operations of the Three-Gorges Dam [Duan *et al.*, 2007; Li *et al.*, 2007; Siswanto *et al.*, 2008]. Previous studies also show the significance of external nutrients imported from southeastern margin, for instance, the Taiwan Strait (TWS) [Chuang, 1986] and the shelf break where the Kuroshio flows northeastward along the Okinawa Trough until it turns eastward through the Tokara Strait [Zhang *et al.*, 2007]. Due to the year-round upwelling, subsurface Kuroshio water northeast of Taiwan carries a considerable amount of low-temperature, high-salinity, and oxygen-unsaturated water into the shelf [Chern *et al.*, 1990; Liu *et al.*, 1992]. In light of the highest concentration of nutrients in the water column, the role of subsurface Kuroshio water has been emphasized [Chen, 1996].

In the southern ECS, the intruding saline Kuroshio water along with the warm water from the TWS forms a mixture on the shelf, known as the Taiwan Warm Current (TWC). It is traditionally thought as the most significant source of heat and salt for the shelf of the ECS [Guan, 1984; Su and Pan, 1987] (Figure 1). Observations always show high-salinity water in the midshelf and less saline water in the inner shelf, which together forms a strong gradient parallel to the coast [Weng and Wang, 1984]. Modeling work has revealed that the nearshore portion of the TWC can approach as far as 30.5°N southeast of the CR mouth in summer [Yang *et al.*, 2012]. Furthermore, the offshore portion of the TWC, together with the Kuroshio intrusion in the northern ECS, has been recognized as the major source of the Tsushima Warm Current (TSWC) [Fang *et al.*, 1991; Isobe, 1999; Guo *et al.*, 2006]. On the other hand, the inner shelf and midshelf water can be delivered

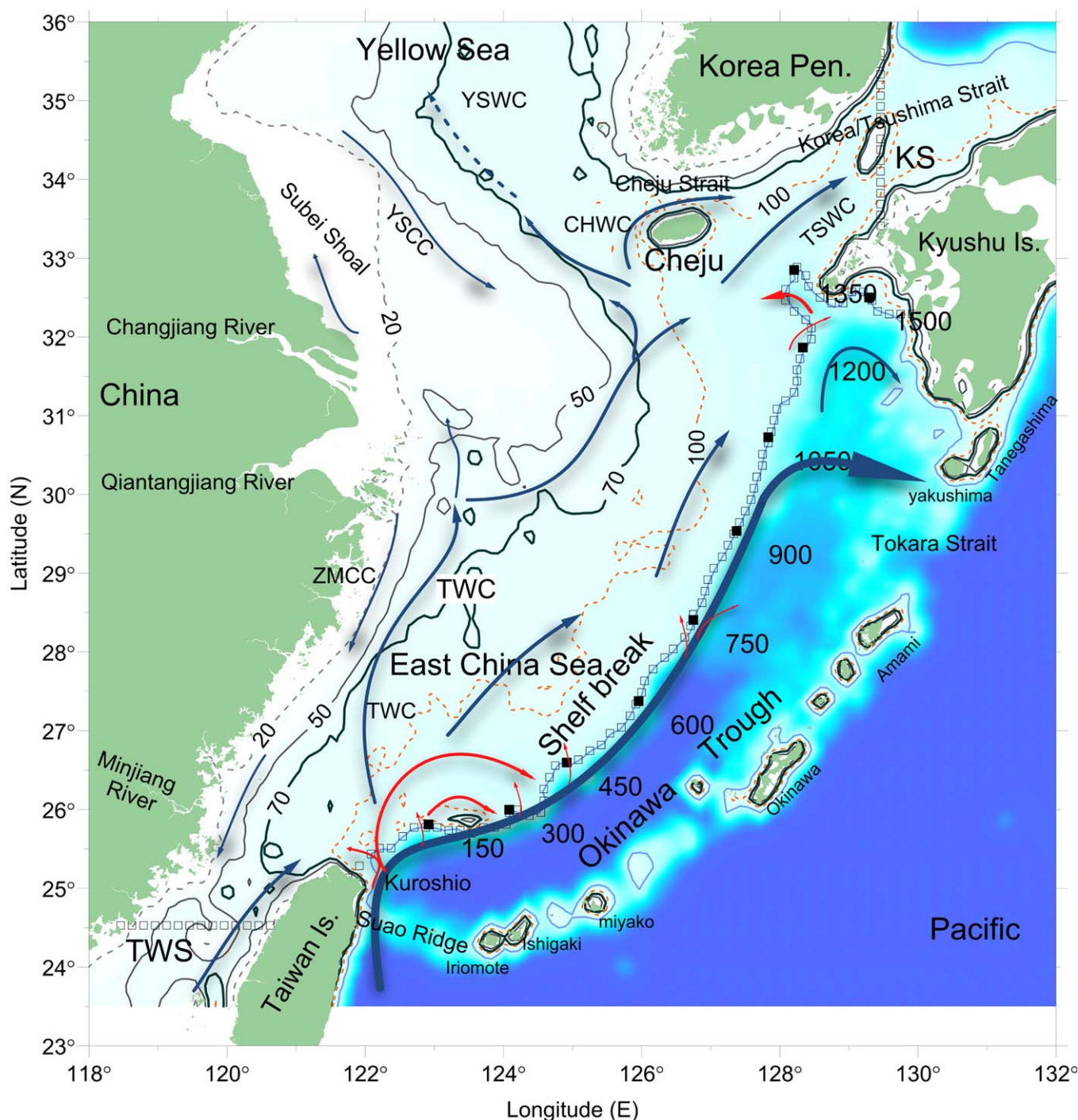


Figure 1. Topographic features, major rivers, and schematic of the circulation in the East China Seas. The arrows indicate the Yellow Sea Warm Current (YSWC), the Yellow Sea Coastal Current (YSCC), the Cheju Warm Current (CHWC), the Tsushima Warm Current (TSWC), the Taiwan Warm Current (TWC), the Zhe-Min Coastal Current (ZMCC), and Kuroshio. The dash arrow indicates the YSWC in winter. The edges (the “fences,” open squares) enclosing the East China Sea shelf consist of the Taiwan Strait (TWS), the shelf break along the 200 m isobath, and two short segments connecting to the Taiwan Island and the Kyushu Island, and the Korea/Tsushima Strait (KS). Places where significant cross-shelf break exchanges occur are indicated by red arrows around the shelf break. Also shown are the distances (100 km between black squares) along the shelf break starting from the Taiwan coast.

to the outer shelf as well, indicated by seaward penetrating of high levels of chlorophyll *a* at the surface [Yuan *et al.*, 2005].

The exchange, in both onshore and offshore directions, is thus obvious and qualitatively recognized by previous studies, highlighting the ECS as a dynamic area for studying the shelf-deep ocean exchanges. The

term “exchange” in this paper means not only the across-boundary volume transport, but also the heat-transport and salt-transport as in *Hsueh et al.* [1997]. A broader definition of “exchange” could also mean transports of nutrients and organic matters, which to some extent have been addressed by previous studies, e.g., *Zhao and Guo* [2011]. However, previous research of cross-shelf exchange is restricted to a qualitative understanding of the underlying process rather than quantitative estimates, which are difficult and disputable due to complicated circulation systems and strong variability [*Liu*, 1998]. Direct measurements are either spatially rare or temporally insufficient to get a full insight into the water budget of the ECS until *Teague et al.* [2003] presented directly measured currents with moorings in the TWS, Korea Strait (KS) and Cheju Strait, while the net cross-shelf-break transport was calculated to close the budget of the ECS, as shown in their Figure 5. On the other hand, specific locations of Kuroshio intrusions along the shelf break were still not determinable with a box-like model [*Teague et al.*, 2003]. Box-like model can neither provide the interior structure and variability of the ECS circulation, nor explain how its edges are connected. It is thus useful and necessary to apply a high-resolution primitive equation model to address these issues. For instance, *Huthnance et al.* [2002] pointed out that the limitation of box study reinforces requirements that models should be used for finer resolution in both space and time, which intrinsically close budgets. *Guo et al.* [2003] suggested that with an improved resolution ($1/18^\circ$), thus better representation of the topography, their model produced more realistic density distribution, and thereby current structure comparing to the low-resolution model ($1/6^\circ$). Facilitated by a high-resolution simulation nested in a large-scale model, *Guo et al.* [2006] presented that the Kuroshio water through the shelf break could be imported at an annual rate of about 1.5 Sv ($1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$), high in fall but low in summer. Their study also indicated that the most substantial exchange occurs northeast of Taiwan, consistent with a persistent upwelling zone there, which is followed by the exchange west of Kyushu. The model of Lee and Matsuno estimated the transport rate across the 200 m isobath at around 2.7 Sv in winter and 2.5 Sv in summer, heavily depending on variations of the through-flow transport in the TWS and the Kuroshio volume transport (KVT) east of Taiwan [*Lee and Matsuno*, 2007]. Based on an 18 year model result, *Song et al.* [2011] found a much stronger seasonality with the exchange in the winter (1.69 Sv) being nearly three times larger than that in the summer (0.48 Sv).

Most of the Kuroshio tends to pass through the continental margin following the contours of constant depth to conserve the potential vorticity. However, the driving mechanism for cross-shelf exchange remains unresolved. *Su and Pan* [1987] suggested that the adjustment of the pressure gradient due to the sudden change of topography northeast of Taiwan is the main reason for the intrusion. It is reported that the seasonal variability of the intrusion northeast of Taiwan is strong during fall to early spring due to cooling along with high pressure over the shelf but weaker in summer related to typhoons [*Chuang and Liang*, 1994]. *Oey et al.* [2010] demonstrated that the onshore intrusion can be enhanced due to the shoreward shift of the Kuroshio northeast of Taiwan in response to surface cooling in winter. Based on 16 year satellite altimetry data, *Liu and Gan* [2012] suggested the shoreward intrusion in the entire vertical water column of the Kuroshio is probably governed by the local geostrophy, while surface transports were mostly affected by Ekman effects. Using an advanced artificial neural network classification algorithm, *Wu et al.* [2014] revealed that strong intrusion northeast of Taiwan occurs more frequently in winter rather than in summer, which is regulated by surface heat flux gradient, wind forcing, and the KVT. *Guo et al.* [2006] suggested the change in density field on the shelf break is the major cause, followed by local wind. *James et al.* [1999] attributed the exchange to the Kuroshio frontal variability, the associated filaments, and cold eddies. The exchange could also be subject to the variability of the KVT east of Taiwan and buoyancy change in the shelf that affects seasonal variations of the Kuroshio position off the northeast Taiwan [*Hsin et al.*, 2013]. *Liu et al.* [2014] presented supportive evidence that the KVT, which is mainly determined in the interior Pacific, is a remote factor of the local Kuroshio onshore intrusion, affecting the latter's volume transport and spatial pattern. The KVT east of Taiwan has been found to be modulated by westward impinging eddies [*Lien et al.*, 2014]. On the other hand, Ekman transport, meandering, and eddies of the main flow, have been shown to facilitate local cross-shelf transport at other ocean margins as suggested by *Huthnance et al.* [2009] and *Zhou et al.* [2014].

As *Qiu et al.* [1990] suggested, most intruding Kuroshio water northeast of Taiwan rejoins the main stream of the Kuroshio. Therefore, it is also worth to know how far exchanges could be delivered in the ECS shelf, since the mixing of the shelf water with the Kuroshio water will modify the properties of shelf water. The northern ECS seems to be vulnerable to eutrophication which results in high productivity and in turn leads to one of the largest hypoxic zone in the world, especially in the recent decade [*Li et al.*, 2002; *Chen et al.*,

2007; Wei *et al.*, 2007; Zhou *et al.*, 2010]. One interesting aspect is that the reported hypoxia extent off the CR in 2006, being one of the largest in the history, occurred mostly north of the CR differing from all previous studies [Zhu *et al.*, 2011]. Impacts of the intruded water have two aspects: first to enhance stratification on the shelf because of its high salinity; second to carry the water with low-concentration of dissolved oxygen into the frequent hypoxic zone off the CR [Zhang *et al.*, 2010].

In addition to the quantification of cross-shelf exchanges along the shelf edge of the ECS, this study aims at understanding how the cross-shelf exchange varies spatially and temporally, particularly at the shelf break, and also at understanding to what extent and how the exchange between the shelf break and the open ocean could be delivered to the nearshore regions, especially for the year 2006 when a significant amount of high salinity water and one of the largest extent of hypoxia were identified north of the CR in the summer. In this study, our focus are the annual cycle and the annual means, as well to what degree the annual cycle and the annual means in 2006 are different from that of the climatological run.

2. Materials and Methods

2.1. Field Data

Hydrographic surveys were carried out on board of R/Vs Beidou and Haijian 49. Three large-scale investigations were conducted off the CR estuary during 2–11 June, 20–30 August, and 3–13 October 2006 (Figure 2). Hypoxia occurred over all three periods [Zhou *et al.*, 2010; Li *et al.*, 2014] and the water off the CR estuary was more saline than normal years. The transection along 31°N was repeated by Haijian 49 on 12 August. Temperature, salinity, and depth were acquired using Seabird CTD.

2.2. Numerical Model

The model was configured based on the Regional Ocean Modelling Systems (ROMS) version 3.7. Details of the ROMS computational algorithms were summarized by Haidvogel *et al.* [2008] and Shchepetkin and McWilliams [2005]. Vertical turbulence closure is the $k-\epsilon$ scheme [Rodi, 1987], which is reformulated by Warner *et al.* [2005] under the generic length-scale framework of Umlauf and Burchard [2003]. ROMS has been employed in the East China Seas [Zhou *et al.*, 2009] and many studies of shelf and deep basin interactions as well [e.g., He *et al.*, 2011; Yang *et al.*, 2011].

The model covers a rectangular area within 23.5°–41°N and 117.5°–132°E, which includes the entire Bohai, Yellow and East China Seas, as well as parts of the northwestern Pacific and East/Japan Sea. The domain has three open boundaries: the southern boundary and part of the eastern boundary are shown in Figure 1, while the northern boundary is northeast of Korea Peninsula at 41°N. The model employs a homogeneous 1/24° horizontal resolution, approximately 3.5 km in the northern domain and 4.2 km in the south, and 30 terrain following vertical layers of s -coordinate system according to Song and Haidvogel [1994]. The bathymetry of this domain varies from a few meters in the inner shelf to several thousand meters in the Pacific. To obtain a finer resolution on the shelf and to reduce the spin-up time, bathymetry off the slope is cut off at 1500 m, resembling that of Chang and Isobe [2003] and Chen *et al.* [2008], which is notably deeper than 800 m and roughly the e-folding square depth ($1/e^2$) of the Kuroshio Current [Yuan *et al.*, 1998; Gilson and Roemmich, 2002]. The configuration is referred to as the BYE24 hereafter.

Tide-induced shear dispersion, not considered in previous studies of Hsueh *et al.* [1997] and Guo *et al.* [2003, 2006], tends to intensify the mixing and to enhance the exchanges. For simplicity, only M_2 tide is considered in the BYE24 model. Tide information is from the TPX07 model [Egbert and Erofeeva, 2002]. We compared two cases driven by the same forcing except that one case includes M_2 tide while the other not. The circulation pattern of these two cases looks rather different throughout the year (not shown). The time step is 92 s, $1/486$ M_2 period, which allows us to remove the tidal signal from the model output easily. The simulation includes also seven major rivers (the Changjiang River, Huanghe River, Liaohe River, Yalvjiang River, Hanjiang River, Qiantangjiang River, and Minjiang River) with the runoff obtained from www.hydrodata.gov.cn and the Datong Hydrological Station, which are similar to those used by Chang and Isobe [2003]. All rivers are provided as monthly fresh water volume discharge, whose weight decreases linearly with the layer number, from 30 at the surface to 1 at the bottom.

The model was run for two experiments: a climatology run with climatological surface forcing and boundary conditions, and a hindcast run for 2006 with real-time surface forcing and boundary conditions. The climatology run was initialized from a climatological mean January derived from the multiyear observations

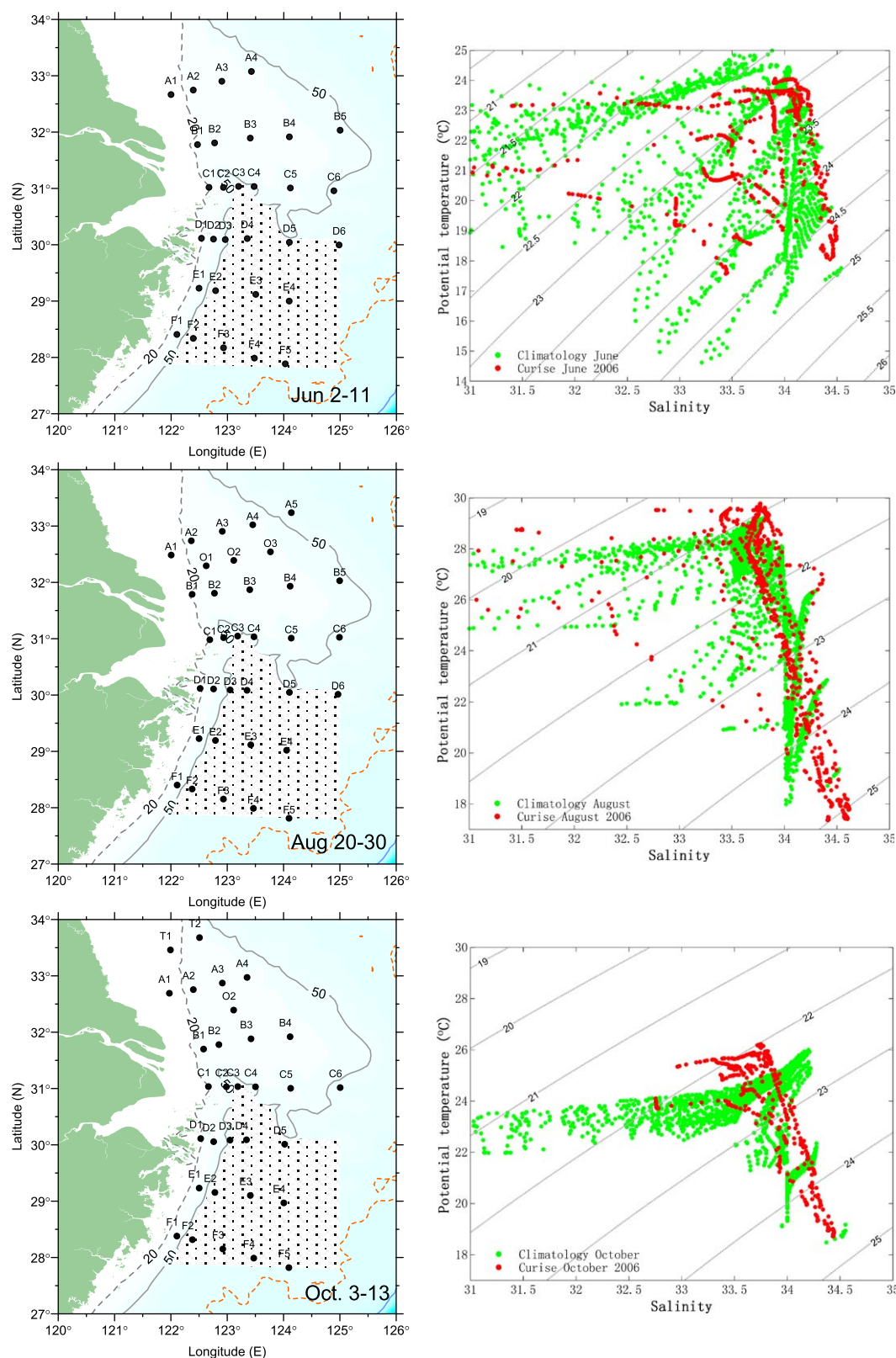


Figure 2. Sampling stations (black dots) for three cruises in 2006 and corresponding temperature-salinity diagrams (red dot) comparing with those of climatology (green dot, based on Chen [1992]). Note that only data measured within the gray zone, where the water depth is deeper than 50 m, and the climatological data in the stippled area are shown in the T-S diagrams.

[Chen, 1992] and run for 10 years driven by half-by-half COADS climatological monthly forcing [Da Silva et al., 1994], which includes net heat flux (Q), surface fresh water flux, wind stress, and short-wave radiation. The advantage of COADS data is its relatively long reference period, which is important in terms of climatological mean. Their reliabilities have been demonstrated by our modeling works for the Bohai Sea and Yellow Seas [Huang et al., 2005; Zhou et al., 2009]. The last 3 years was averaged as the climatological mean. Temperature, salinity, and current from the December of the last year were extracted as the initial condition for the hindcast run of 2006. Though it is relatively long to adjust temperature and salinity to 2006, the response of circulation and surface elevations to wind is short. On the other hand, the initialization from winter when temperature and salinity are vertically well-mixed should also reduce the influence of initial conditions. This type of spin up for a simulation under realistic forcing was commonly used, e.g., Liu et al. [2010] and Zhou et al. [2014]. The hindcast run was driven by meteorological forcing and boundary conditions for 2006. The momentum flux was from the Blended Sea Winds, which provide 0.25° gridded ocean surface vector winds [Zhang et al., 2006]. Wind stress was calculated using the Large and Pond formulation [Large and Pond, 1981]. Missing records of the Blended Sea Winds were interpolated or extrapolated with a robust technique based on a penalized least squares method [Garcia, 2010]. The hindcast model also used the direct surface heat and fresh water fluxes from the 3 hourly ECMWF ERA-Interim [Dee et al., 2011]. Boundary conditions such as the temperature, salinity, velocity, and sea level elevation come from HYCOM simulation for 2006 [Cooper and Haines, 1996; Cummings and Div, 2006].

A heat flux correction with respect to the surface temperature, $dQ/dSST$, was derived from bulk formulas and used to introduce thermal feedback with a time scale of 30 days [Marchesiello et al., 2001]. Monthly sea surface salinity (SSS) from WOA was applied in both experiments to correct the sea surface fresh water flux with the scheme suggested by Barnier et al. [1995]. The model grid, meteorological data set, rivers, and tide were prepared with ROMS_tools [Penven et al., 2008], which was further adjusted for the ECS configuration [Zhou et al., 2009]. Both the climatological run and 2006 hindcast outputted temperature, salinity, and current fields averaged over every consecutive six M_2 periods.

2.3. Calculation of Cross-Shelf Transport

The volumetric transport of water between the shelf of the ECS and deep-ocean regions was calculated along an enclosed edge consisted of three “fences” (Figure 1, lines of open squares). The fence along the shelf break (the shelf break fence) approximates the 200 m isobath plus two short segments connecting to the Taiwan Island (TI) and Kyushu Island (KI). The fence in the TWS (TWS fence) is at $24.5^\circ N$ and the fence in the Korea Strait (KS fence) is along $129.5^\circ E$. For the climatological run, the transport was calculated based on a quasi-monthly mean (60 M_2 periods) model results, while for the hindcast run, the total transport was calculated each six M_2 periods.

3. Results

3.1. Saline Water off the CR Estuary

Salinity is a good tracer which is able to identify different water masses between the inner shelf and the outer shelf and their cross-shelf movements. The low-salinity water originates from the inner shelf while the saline water mostly comes from the TWS and the shelf break. Figure 2 shows the survey stations during three cruises and temperature-salinity diagrams for the data sampled at stations deeper than 50 m where it is usually thought to be the main northward passage of the saline and warm TWC. As a mean state, diagrams from climatological atlas are shown as well, and they cover a slightly wider spatial region (shaded gray in the left plots). The TWC spreads northward, of which the portion under the thermocline is featured by high salinity ($34.2\text{--}34.6$) and low temperature ($17^\circ C\text{--}23^\circ C$), while the portion in the mixed layer has more wider range of salinity ($33.3\text{--}34.2$) and temperature ($23^\circ C\text{--}29^\circ C$) in the southern ECS [Weng and Wang, 1984]. To account for variations of the TWC with latitude, our definition of the TWC water is slightly different by using isohalines of 34.0 and 34.5 for the bottom water, and 32.0 or 33.0 for the surface water. Figure 2 shows similarities of TWC between data from climatological atlas and measurements in 2006. It is noted that both salinity at station F2 and F5 are higher than their ambient stations (not shown), inferring different water mass merging into the TWC south of $28^\circ N$. In August, in noticeable contrast to June, measurements in 2006 are more saline than the climatological mean with the maximum salinity as high as 34.6 at F3 and F5, followed by F2 and F4 where the salinity is all larger than 34.5. Comparatively, the central

position of the TWC has shifted gradually offshore further than that in June. The unusually high-salinity water can also be found in October when temperature decreased significantly below 26°C.

Further evidence is provided with measurements from these surveys along with a repeated investigation in August at 31°N, typically the northern end of the TWC in summer [Yang *et al.*, 2011]. Figure 3 shows the evolution of temperature and salinity during these periods. Temperature profile in June shows two cold domes below the thermocline. One was on the western edge of the submerged valley (deeper than 50 m), and the other was at the eastern end of the transect, colder than the western one. Documented by previous studies, the western cold water was related to the TWC in view of its high salinity of 33.5 [Chen *et al.*, 2003]. The cold water on the right end of the transect does not seem to be a part of the TWC owing to its horizontal distribution (not shown) and less saline attribution. In other words, the TWC was not strong enough to approach 31°N in June. In mid-August, the salinity of the bottom water increased by 0.5 despite more fresh water nearshore. A doming of the 33.5 salinity contour was recognized at the surface above the valley, and it was manifested as a strong doming of thermocline as well. Patches of fresh water with the salinity less than 33 appeared to be detached from the inner shelf front implying significant upwelling. The repeated survey along the transect in late August demonstrated a similar pattern with enlarged coverage of high-salinity water, but the doming of the cold bottom water was suppressed. The striking feature in August is that the saline water with salinity higher than 34.0 appeared near 125°E in addition to the valley at this transect, which is rare and implies the existence of offshore branch of the TWC at this place. The intensified northward intrusion of the offshore branch of the TWC will be illustrated further in the following paragraphs. The TWC retreated in October with only the nearshore branch found above the valley while the offshore branch disappeared.

3.2. Spreading of the CR Plume and Saline Water Intrusion

Among three large-scale surveys in 2006, observations in August were also manifested by quick changes of nearshore plume water and striking doming of saline water (Figure 3). Sea surface salinity, therefore, contains rich information of how cross-shelf exchanges occur in August (Figure 4a). First, the spreading of low-salinity water off the inner shelf can be identified from the field data. This cruise, starting from the north to the south, has revealed two significant seaward fresh water tongues. The first signal was detected during 20–21 August, when the fresh water from the CR was transported to the north of the investigating region between 32°N and 33°N and part of it turned eastward and southward. The minimum salinity is 27.37 centered at station A5 (124.072°E, 33.183°N). A few days later, one more low-salinity tongue appeared around 29°N–30°N along the northern Zhejiang coast, approaching the 70 m isobath. This block of low salinity waters, resembling the earlier one, can also be traced back to the CR. Concurrent with the offshore transport of these low-salinity waters, a saline water tongue with salinity higher than 33 appeared on the midshelf, intruding northeastward off the northern Zhejiang coast with a part of it penetrating into the 50 m isobath and turning northwestward. The western tip of the saline tongue was only about 150 km away from the mouth of the CR.

While having provided many valuable insights into how strong northward intrusion occurred, observations were limited in both temporal and spatial coverage. Simulation and corresponding analysis were thus carried out to show how the intrusion occurred. The hindcast run reproduced the two blocks of low-salinity waters in a reasonable fashion in terms of both their spatial and temporal coverage (Figures 4b–4d). Furthermore, it replicated a clear pattern change in the CR diluted water, which spread eastward first and then veered to the southeast quickly during the survey periods. It is inferred from the model that the northern low-salinity water in Figure 4a was actually a detachment from the CR plume. The salinity within the detached patch increased gradually due to mixing as it was transported to the east and south. Low-salinity water detachment from the river plume is quite often observed and more observations have been presented from 2001 to 2006 and even earlier periods, for instance, Chen *et al.* [1994, Figure 6], Lie *et al.* [2003, Figure 4c], Moon *et al.* [2009, Figure 2], and Xuan *et al.* [2012, Figure 1]. The detachment of plume water could be determined by baroclinic instability of the river plume and eddy formation [Chen *et al.*, 2008] or Ekman effects associated with southerly monsoon [Xuan *et al.*, 2012], which has also been discussed by Moon *et al.* [2010] and Wu *et al.* [2011].

The southern block of the low-salinity water, connecting with the major part of the plume, resulted from an altered extension of the CR diluted water. The southeastward expansion of the diluted water, which is

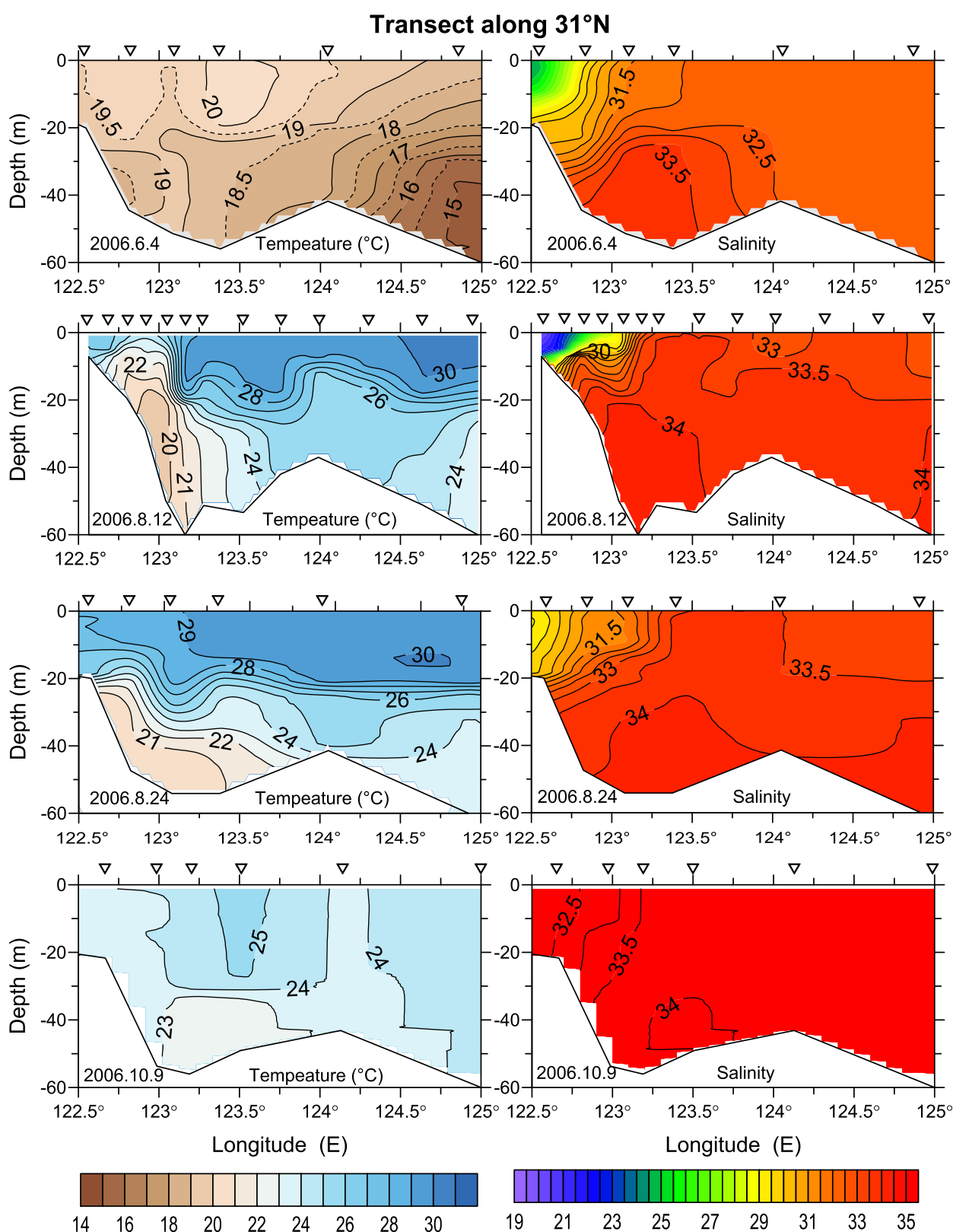


Figure 3. Observations of temperature and salinity along the 31°N transect in the northern East China Sea. During summer-fall 2006, rarely seen saline water with salinity higher than 34.0 was observed near the seabed at the transect. Contour interval (solid line) is 1°C for temperature and 0.5 for salinity. Sampling stations are marked by down triangles above the transect.

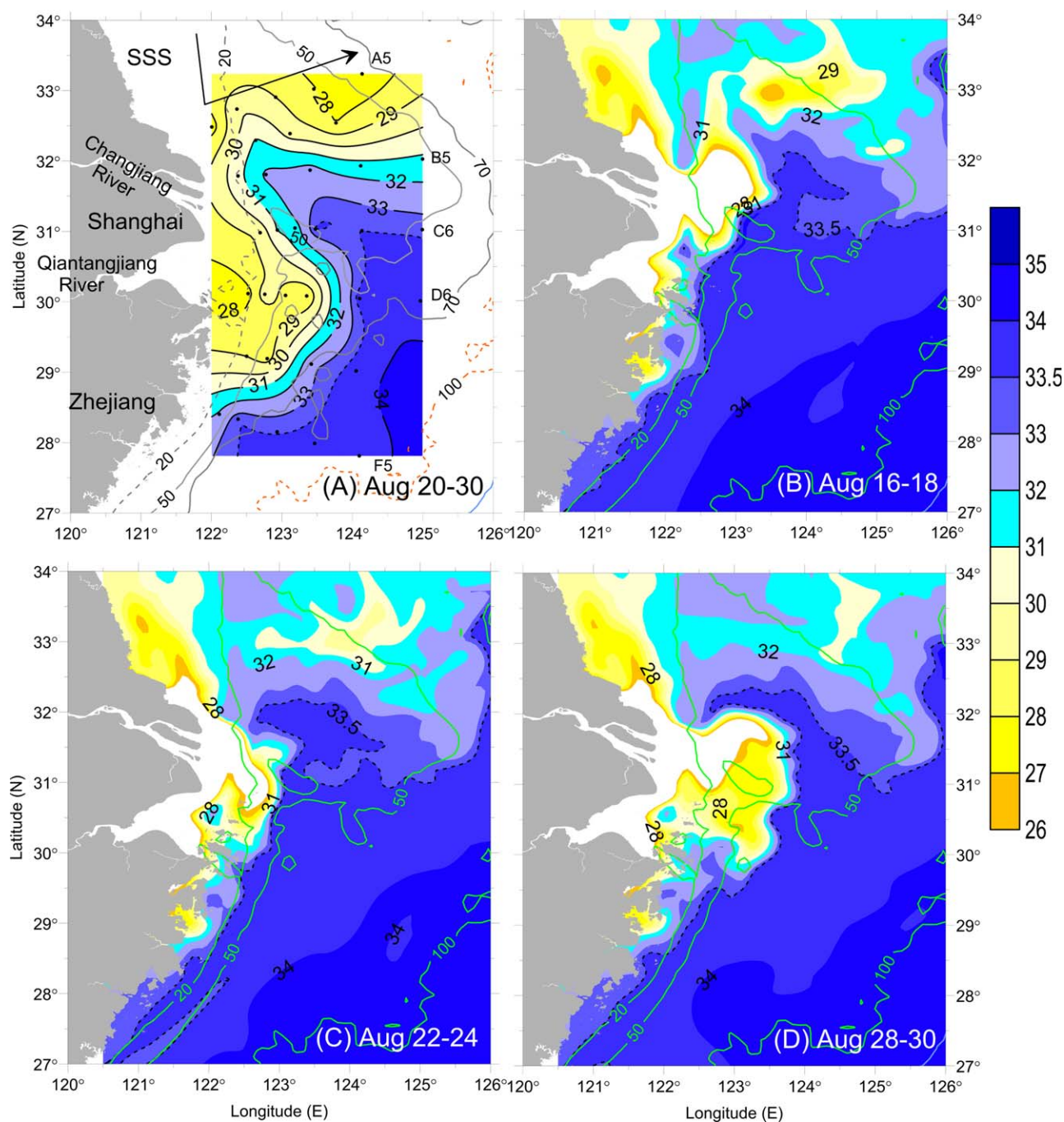


Figure 4. Comparison of the sea surface salinity between the hindcast and in situ measurements in 2006. Shown in (a) are observations acquired from August 20 through 30 in 2006 [after Zhou et al. 2010]. Samplings are marked by dots started from the north indicated by the arrow. Shown in (b–d) are 3 day-mean modeled SSS. Salinity less than 26 is shaded as white and contour of 33.5 shown as dashed line. Isobaths of 20, 50, 70, and 100 m are shown in gray lines.

usually seen in autumn but rarely in summer, could also be a significant contributor to the cross-shelf exchange as it crossed the 70 m isobath. The modeled quick change in the extension of the river plume in the form of a southeastward bulge is consistent with field measurements.

As discernable from field data, a strong incursion of high-salinity water with the surface salinity from 32 to 33, arriving at the north of the submerged river valley indicated by the 50 m isobath, was also visible from the model. Although it is common for the high-salinity water occurring to the southeast of the river mouth in summer due to the intrusion of the TWC favored by the southeasterly monsoon [Yang et al., 2013], the

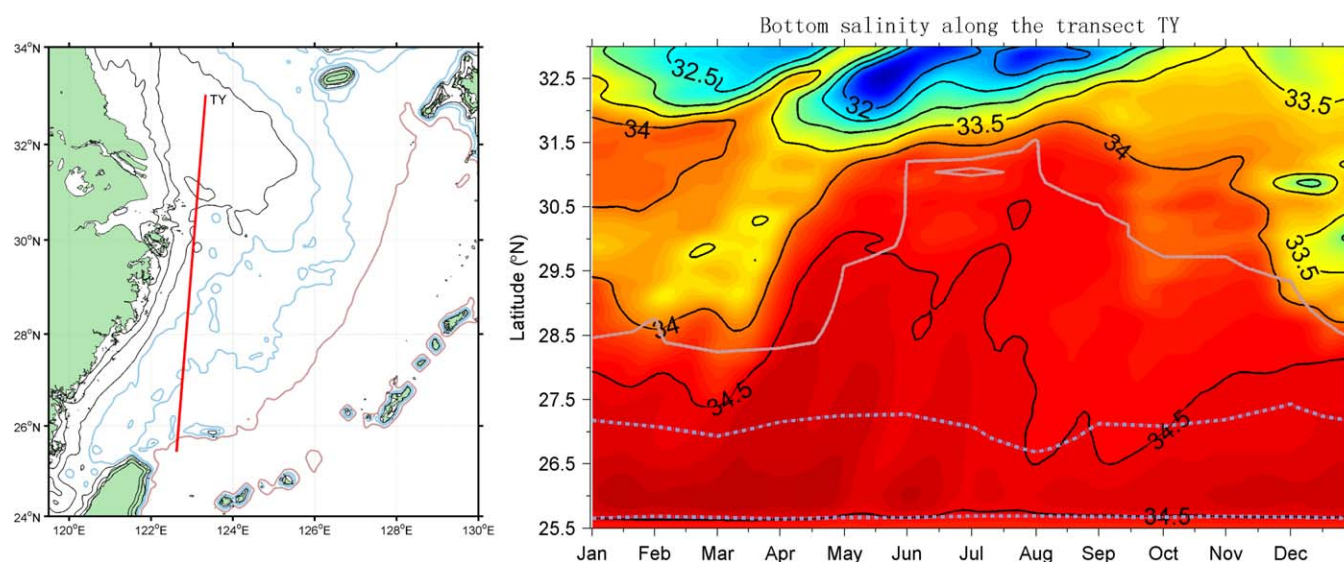


Figure 5. Hovmöller diagram of the modeled salinity (shaded, black solid contours) at seabed along the transect TY (from Taiwan to the Yellow Sea as shown in the upper plot) from the hindcast run. Contour interval is 0.5. The climatological mean bottom salinity of 34.5 (gray dash-dot line) and 34 (gray solid line) digitized from Chen [1992] are shown as well.

intrusion of the TWC in 2006 was comparatively strong. More specifically, the surface isohaline of 32 reached 32°N in August 2006, in contrast to the usual 31°N in normal summers [Lie *et al.*, 2003, Figure 2]. The minimum distance of the modeled saline water tongue to the river mouth was no more than 150 km, rather similar to the observed.

To illustrate the intrusion of the TWC in 2006 comparing with climatology, bottom salinity along a transect from north of the TI to the Yellow Sea (transect TY) was extracted from the hindcast run. Its Hovmöller diagram [Hovmöller, 1949] along with its counterpart from the climatological run was shown in Figure 5. The climatological mean suggests the TWC remains weak till late April as indicated by the 34 isohaline (dashed red line). The TWC is then enhanced during the transition period in April–May and further reaches its strongest in June–August when it is pushed northward as far as 31°N by the summer monsoon. In the hindcast, the TWC was also weak in cold seasons as in the climatology data but slightly more northward centered around 28°N–29°N. The strengthening of the TWC was 1 month earlier and lasted 2 months longer than the climatology. The TWC reached the farthest to 32°N in mid-August (34 isohaline), which was about 2 weeks later than usual. Frequent low-salinity patches were discernable in the shelf from both hindcast and climatological data, which mostly occurred between 29°N and 30°N from December–April.

In contrast, the 34.5 isohaline tells a rather significant seasonal variability. Climatological data show the high-salinity water frequently approaches 27°N and it can retreat occasionally to the south of 26°N, for instance, first in February, and again during late July to early October. A probable reason is that the TWC water consists partly of the low-salinity water from the South China Sea through the TWS, which is relatively low in salinity in comparison to the Kuroshio especially in summer [Jan *et al.*, 2006]. It is also noted that another portion of the TWC water coming from the intruded Kuroshio water northeast of Taiwan is subject to tropical storms [Chern *et al.*, 1990] and storm-induced precipitation tends to lower the salinity. However, the reduction of salinity in February remains unclear. The model shows consistent TWC seasonality with observations, which has high salinity in winter from December to February but mostly confined to the south of 28°N. The 34.5 isohaline was pushed northward to 29.5°N from May to June. It similarly disappeared from late June to mid-September, but appeared again and reached 28°N in December. Additionally, it is apparent that the TWC, inferred from both its surface (Figure 4) and bottom (Figure 5) signals, had a strong intrusion in 2006 probably associated with considerable cross-shelf exchanges, which were further delivered to the north of the CR mouth. It is also noted that the intrusion in the shelf has a time lag when it proceeds northward, for instance, the intrusion peaked in May and June in the southern ECS, while reached the maximum around August off the CR, indicating an average speed of 0.15 m s^{-1} .

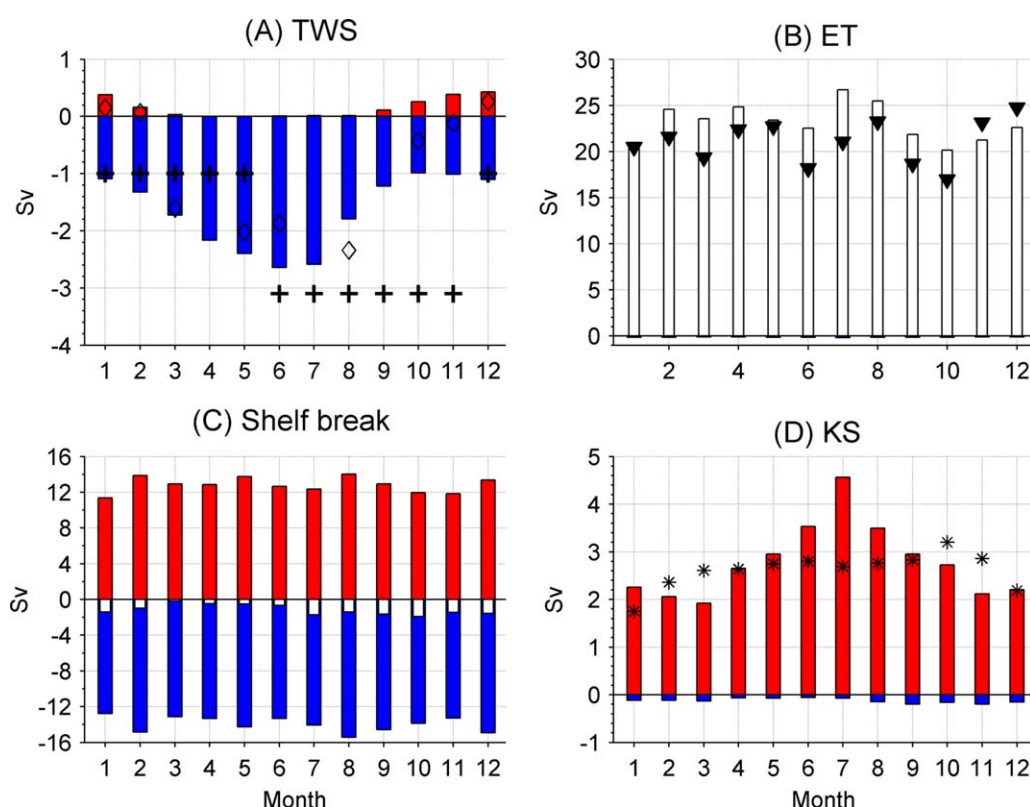


Figure 6. Monthly volume transport through the TWS (a), the transect east of Taiwan (ET, b), the shelf break (200 m isobath, c), and the Korea Strait/Tsushima Strait (KS, d) from the climatological run. Outgoing (flowing out of the shelf of the East China Sea) flow is positive (red), incoming flow is negative (blue), net is white, except for B, where the transport is positive for northward. Net transport measurements reported by Fang et al. [1991] (plus), Jan et al. [2006] (diamond), Lee et al. [2001] (solid inverted triangle), and Takikawa et al. [2005] (asterisk) are shown as well.

3.3. Boundary Exchanges of the ECS

The exchange between the inner shelf and midshelf in the northern ECS has been demonstrated based on hydrographic surveys in the last sections. High-salinity water in the ECS can only come from the open ocean through the TWS or the shelf break as investigated in previous studies [Guo et al., 2006; Isobe and Beardsley, 2006; Yang et al., 2012]. This section will illustrate how much and where deep-basin water can be transported onto the shelf. Transports crossing three edges, i.e., TWS, KS, and the shelf break fences (see Figure 1), of the ECS shelf were calculated from the climatological run and hindcast run, respectively. For clarity, flows across any of the three fences into the ECS were defined as inward (negative), while those leaving the area defined as outward (positive).

Figure 6 shows volume transports through the three fences and east of Taiwan from the climatological run. TWS mostly contributes inward (northward) flow to the ECS with an annual mean of -1.67 Sv, while the outward flow only accounts for a small percentage mostly in Fall (September–November) and winter (December–next February) with an annual mean of 0.15 Sv (Figure 6a). The monthly net transport through the TWS ranges from -0.62 to -2.64 Sv with a standard deviation of 0.78 Sv (Table 1). These seasonal or annual mean values are consistent with the results derived from previous observations by Fang et al. [1991] and Jan et al. [2006]. The weak but highly varied northward transport through the TWS in winter is in consistency with the mooring observations presented by Lin et al. [2005] and Teague et al. [2003] (Figure 7b).

The Kuroshio, a major source of saline water onto the ECS, whose volume transport is usually confined within a narrow band east of Taiwan before impinging onto the shelf break [Hsin et al., 2008; Wu et al., 2014]. We calculated the KVT along a transect near Suao Ridge from Taiwan to 123°E at 24.5°N (see Figure 1). The maximum net transport of 24.89 Sv appears in summer, followed by spring (23.92 Sv), and winter (22.55 Sv), and the minimum in fall (21.08 Sv). The estimates resembles the results derived from mooring

Table 1. Mean Values and Standard Deviations of Volume Transports Crossing ECS Boundaries From the Climatology Run Based on Monthly Output^a

Fence Name	Outward	Inward	Net
TWS	0.15	−1.67	−1.52 ± 0.78
KS	2.79	−0.12	2.67 ± 0.80
Shelf break	12.82	−13.99	−1.17 ± 0.56
Sum	15.76	−15.78	−0.02 ± 0.14

^aThe fences are indicated as open squares in Figure 1. Positive transport is directed outward. Unit: Sv (1 Sv ≡ 10⁶ m³ s^{−1}).

summer at a mean rate of −1.32 Sv and −1.26 Sv. Due to the largest outward transport and the weakest inward counterpart in the year, the net contribution is only −0.40 Sv in spring. Monthly mean exchange is essentially shoreward (Table 1) with an annual average of −1.17 Sv, which is within reasonable range derived from previous model studies (Table 4).

The KS is the most important export in light of net exchange since the ECS receives external waters from the other two edges. The shelf water moves out with a mean value of 2.79 Sv, more than 20 times of the incoming water (Figure 6d). The net value is very close to those derived from mooring observations (2.65 Sv) [Teague *et al.*, 2005] and shipboard ADCPs [Takikawa *et al.*, 2005; Ostrovskii *et al.*, 2009].

The mean transports in and out of the ECS through the three fences from the climatological run are summarized in Table 1. The computational error of the net volume transport is almost negligible, considering riverine fresh water discharge (0.034 Sv) and net precipitation on the sea (0.009 Sv). Comparatively, the shelf break is the major passage where the shelf and the Kuroshio interact, but the largest contributor to the net exchange is from the TWS. Large standard deviations shown in Table 1 represent significant temporal variations.

Similar to the above calculation, transports through the fences and their overall contributions for 2006 are shown in Figure 7. In terms of seasonal mean, the TWS had the largest inward exchange in the summer and the second in the spring, with the largest outward flow in the winter. The net exchange mainly occurred in the summer and in the spring. By contrast, the shelf break approached the largest onshore transport in the summer and the largest offshore in spring, with the largest net exchange occurring in the fall followed by the summer. For simplicity, the annual mean exchanges through the three boundaries are listed in Table 2 and the computational error is less than 1%. Comparisons with Table 1 show that the outward transport through the TWS was again negligible on average, while the inward portion rose to −1.93 Sv, roughly 16% larger in 2006 than normal years. The riverine load and net precipitation is about 0.05 Sv, similar to the climatological value. The striking feature is that the exchange through the shelf break increased tremendously, for instance, the annual mean outward and inward in 2006 increased by 27% and 36% than the climatological ones. As a result, the net transport was −2.59 Sv, up 120% on the climatological run but still comparable to values suggested by previous studies (Table 4). Although the net exchange through both the TWS and the shelf break increased compared to the climatological mean, the transport through the shelf break exceeds that through the TWS by 40%, differing from the climatological run. The overall net flow through the TWS and the shelf break was at a rate of −4.42 Sv for 2006, which was nearly 140 times of the discharge from the CR in the year. Similarly, the net transport through the TWS and the shelf break shows large seasonality in both runs (Figures 6 and 7).

The amount of heat and salt crossing the fences of the ECS are both proportional to the volume transport for the year, with their annual mean values shown in Table 3. The annual mean in and out heat transports through the boundaries are both around 1.9×10^{15} W with a net inward rate of 0.05×10^{15} W. The values are in a reasonable agreement with those presented by Fang *et al.* [2003], considering that the exchange occurred at different fences, i.e., inflow 2.32×10^{15} W and outflow 2.11×10^{15} W (their Figure 3b). Similarly, the inward salt transport is slightly larger than the outward counterpart with difference being 0.05×10^8 kg/s, which could be partly balanced by net freshwater input to the ECS, such as, the net precipitation and river discharge.

The intensified exchange through the shelf break and the TWS is related to the Kuroshio. To illustrate the Kuroshio contribution, we compared the KVT across Suao Ridge in 2006 with that from the climatological

observations that peaked at 24 Sv in summer but lowered to 20 Sv in fall [Lee *et al.*, 2001], which yielded the overall mean transport of 21.5 Sv with an uncertainty of 2.5 Sv [Johns *et al.*, 2001].

The exchange across the shelf break is slightly more active in summer in manners of large amount of inward and outward transport (Figure 6c). However, the net volume contribution reaches the largest during fall with a seasonal mean of −1.66 Sv, followed by winter and

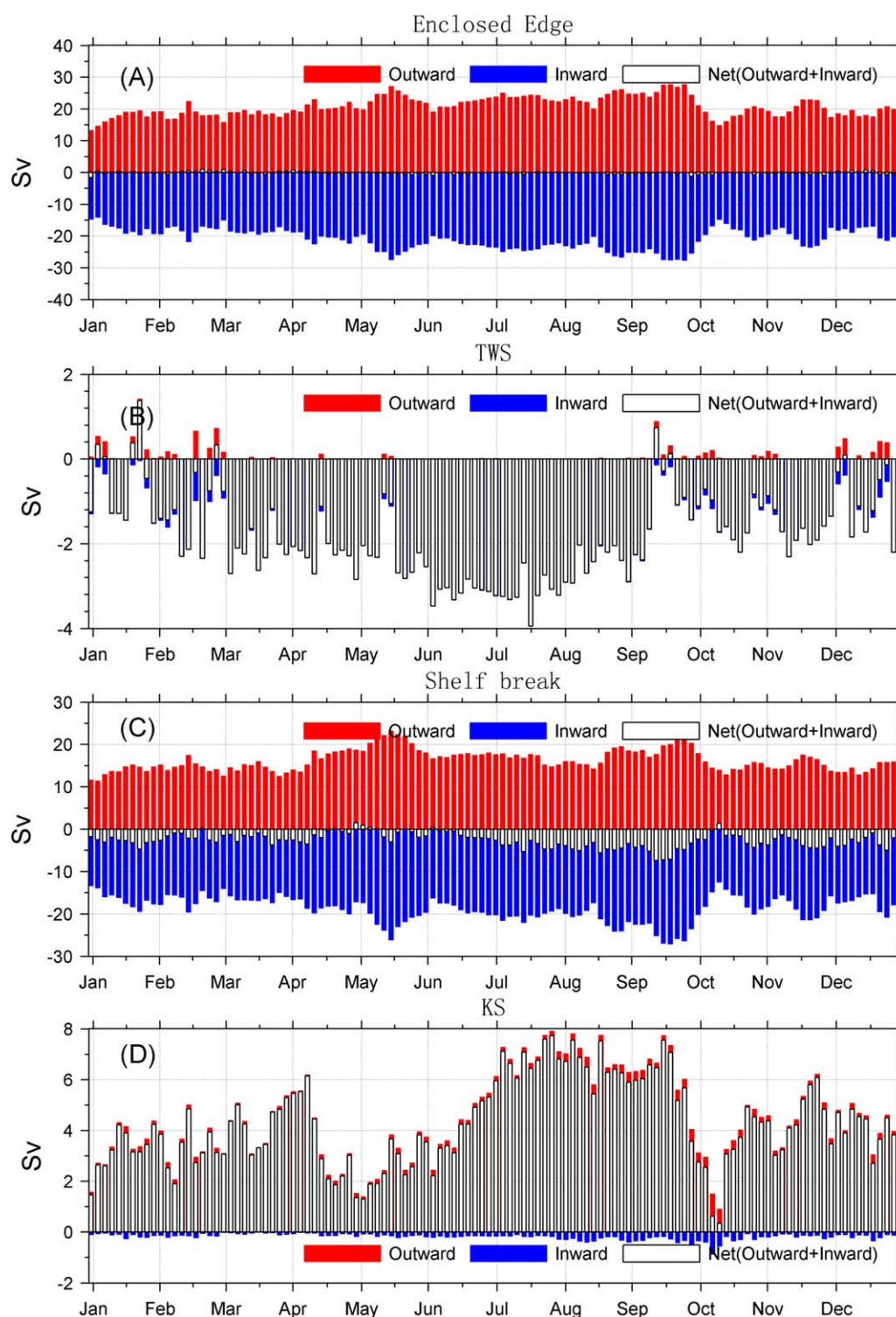


Figure 7. Volume transport (a–d) across the enclosed edge (A, E, and I) of the ECS shelf. The transport is vertically integrated from the surface to the seabed from the hindcast run.

run (Figure 8). The annual mean KVT was 27.5 Sv in 2006 in contrast to the climatological mean of 23.1 Sv. Moreover, the KVT in the hindcast showed different seasonality. It was 31.3 Sv in the spring, second largest, decreased to 17.8 Sv in the summer, rose again to the maximum value of 31.9 Sv in the fall, and then lowered to the winter mean of 28.9 Sv. Both experiments imply that exchanges peak in fall. The rapid reduction

Table 2. Mean Values and Standard Deviations of Volume Transports Crossing Fences of the ECSs During 2006 for the Hindcast Run Based on the Output From Every Six M_2 Cycles (Unit: Sv)

Fence Name	Outward	Inward	Net
TWS	0.09	−1.93	−1.83 ± 1.03
KS	4.51	−0.20	4.31 ± 1.69
Shelf break	16.34	−18.96	−2.59 ± 1.73
Sum	20.94	−21.09	−0.11 ± 0.42

of the KVT in the summer 2006 probably resulted from tight passages of typhoon [Zheng *et al.*, 2014], for instance, super typhoon Chanchu (8–18 May), super typhoon Ewiniar (29 June to 10 July), tropical storm Bilis (9–14 July), typhoon Kaemi (22–25 July), and super typhoon Saomai (4–10 August) (<http://weather.unisys.com/hurricane/index.php>). It was also recognized that impinge of westward anticyclonic eddies may account for

the change in the KVT east of Taiwan [Yang *et al.*, 1999; Zhang *et al.*, 2001]. Details of the related variability and mechanism have been introduced in Lee *et al.* [2013], hence are not included in this study.

Based on above calculation, the annual water budget of the ECS shelf from the climatological run (not shown) and the hindcast run is summarized and the latter is shown in Figure 9. From the annual mean perspective, the riverine load and net precipitation was approximately 1/30 of that through the TWS and 1/50 of that through the shelf break. The exchange through the shelf break varied spatially, with significant inflows northeast of Taiwan, at 26°N, at 28°N and in the northern end of the Okinawa Trough (32.5°N), and with major outflows around 26°N, around 27°N and south of 28.5°N. Results of the budget contain small errors due to the staggered model grid and slight unsteadiness of the momentum, but comparisons with the fluxes from previous studies are instructive that the cross-shelf exchange is all significant (Table 4).

Figure 10 further shows the temporal and spatial variations of the vertically integrated transport through the shelf break. The major swap passages persisted most of the year and hardly changed except that north-

east of Taiwan, which shifted northward from May to August. On the other hand, the seasonality of the cross-shelf exchange at these places is different from each other. The onshore transport is stronger with the location closer to the TI in fall and winter, but weaker in spring and summer when it is also found to shift northward from May to August. Around 26°N and 28°N, cross-shelf exchanges are dynamic in both directions, which are both stronger in spring and summer. In the northern end of the Okinawa Trough (32.5°N), the onshore intrusion is much stronger in early autumn.

To illustrate the further movement of the intruded water on the shelf, one more fence was extracted along the offshore edge of the midshelf, roughly along the 100 m isobath and plus a short segment connected to the Korea Peninsula (Figure 9). After the slope water was delivered to the offshore edge of the midshelf, the two-way transport was still very large with a total of inward rate of 7.55 Sv and outward rate of 8.68 Sv south of the Cheju Island, leading to a net outward transport of 1.13 Sv. Comparatively, the total inward water (7.55 Sv) through the outer shelf into the midshelf is nearly four times of that from the TWS. It is apparent that the southern ECS is

Table 3. Mean Heat (10^{15} w) and Salt (10^8 kg s^{-1}) Transport Crossing Fences of the ECS During 2006 for the Hindcast^a

Fence Name		Outward	Inward	Net
TWS	HT	0.01	−0.21	−0.20 ± 0.12
	ST	0.03	−0.67	−0.64 ± 0.36
KS	HT	0.36	−0.02	0.34 ± 0.18
	ST	1.58	−0.07	1.51 ± 0.59
Shelf break	HT	1.48	−1.67	−0.19 ± 0.14
	ST	5.79	−6.71	−0.92 ± 0.61
Sum	HT	1.85	−1.90	−0.05 ± 0.07
	ST	7.40	−7.45	−0.05 ± 0.15

^aHeat transport and salt transport are referred to as HT and ST for short with positive values representing outward transports.

Table 4. Comparison of the Net Transport Through the Shelf Break with Previous Studies (unit: Sv)

Period	Transport	Source
Climatology	−2.2	Fang <i>et al.</i> [2003] ^a
1999–2000	−3.0	Teague <i>et al.</i> [2003] ^b
1991–1998	−1.46	Guo <i>et al.</i> [2006]
Climatology	−0.85	Isobe and Beardsley [2006]
Climatology	−1.4	Isobe [2008]
Climatology	−2.61	Lee and Matsuno [2007] ^c
1995–1999	−0.93	Liu <i>et al.</i> [2010]
1993–2010	−0.72	Liu <i>et al.</i> [2014] ^d
1991–2008	−1.18	Song <i>et al.</i> [2011]
Climatology	−1.17	This study
2006	−2.59	This study

^aThe transport was calculated based a fence from the TI through Ryukyu Islands to KI instead of 200 m isobath.

^bThe net volume transport through the shelf break was estimated to balance the mass flux in the ECS.

^cAverage derived from summer mean transport of −2.47 Sv and winter counterpart of −2.74 Sv.

^dThe shoreward transport of 0.87 Sv across the 200 m isobaths was subtracted by the southward transport of 0.15 Sv west of Kyushu.

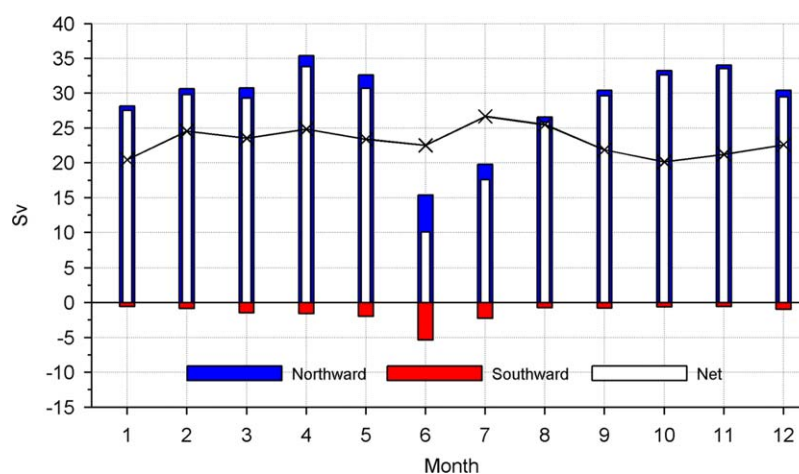


Figure 8. Volume transport across the 24.5°N transect (Suao Ridge) east of Taiwan in 2006 (bar) in comparison with that the climatological mean (line with cross). The transport was calculated from the coastline of Taiwan to 123°E, and the annual mean net transport is 27.5 Sv from the hindcast run versus 23.1 Sv from the climatological run.

featured by vigorous mixing between the TWS water and the cross-shelf water. The annual mean depth-integrated current field (gray vectors) shows that most mixed water spread northward within the midshelf, while a portion of it returns to the outer shelf steered by the bathymetry, especially from 27°–28.3°N. A nearshore portion of the TWC could also be recognized at regions shallower than 50 m in the southern ECS. The TWC bifurcated from 29° to 30.5°N and separated into an onshore branch and an offshore branch. The onshore branch, though much weaker, enters the submerged river valley, and some of it could penetrate across the 50 m isobath and reach 32°N. The stronger offshore branch turned eastward and northward, with 0.79 Sv flowing eastward through the Cheju Strait and then entering the KS.

The annual mean volume flux was derived to illustrate the vertical structure of the cross-shelf exchange from the hindcast (Figure 11) and the climatological run (not shown). The volumetric flux was essentially barotropic, indicated by its nearly homogeneous distribution from the surface to the seabed. The baroclinic signals, with larger values in upper and middle layers but smaller values in lower layers, are more pronounced in the southern part of the shelf break. Dynamic exchanges occurred only at certain places that have been depicted in Figures 9 and 10.

4. Discussion

The Kuroshio water and the TWS water have long been considered important sources of saline water in the shelf of the ECS. Previous studies often show signals of the TWC southeast of the CR mouth, especially in summer. In this paper, observational evidence of the TWC reaching north of the CR mouth in 2006 is presented. It is attributed to stronger than typical cross-shelf exchanges at the shelf break and flows through the TWS, which is reproduced in a 3-D high-resolution model simulation. Moreover, the model provides a quantitative understanding of cross-shelf exchange in the ECS and their corresponding mechanisms.

4.1. Spatial Distribution of the Annual Mean Cross-Shelf Exchange

Several places at the shelf break are identified as persistent and vigorous exchange zones where large amount of onshore and offshore transports occur throughout a year. Except for northeast of Taiwan and in the northern end of the Okinawa Trough (32.5°N), areas around 26°N and around 28°N are newly reported, which are characterized by nearly year-round cross-shelf exchanges as shown in Figures 9 and 10. Similar patterns were also found from the climatological run. These two newly found important exchange zones could be supported by trajectories of eight surface drifting buoys (Figure 12) that were released from 2002 to 2013 by Japan Meteorological Agency. The role of these two places has not been discussed except that around 28°N where the magnitude of onshore exchange was reported stronger than the offshore counterpart [Song *et al.*, 2011].

Importance of the shelf break northeast of Taiwan to the cross-shelf exchange has been demonstrated by both field data [Chern *et al.*, 1990; Liu *et al.*, 1992] and simulations [Guo *et al.*, 2006; Oey *et al.*, 2010;

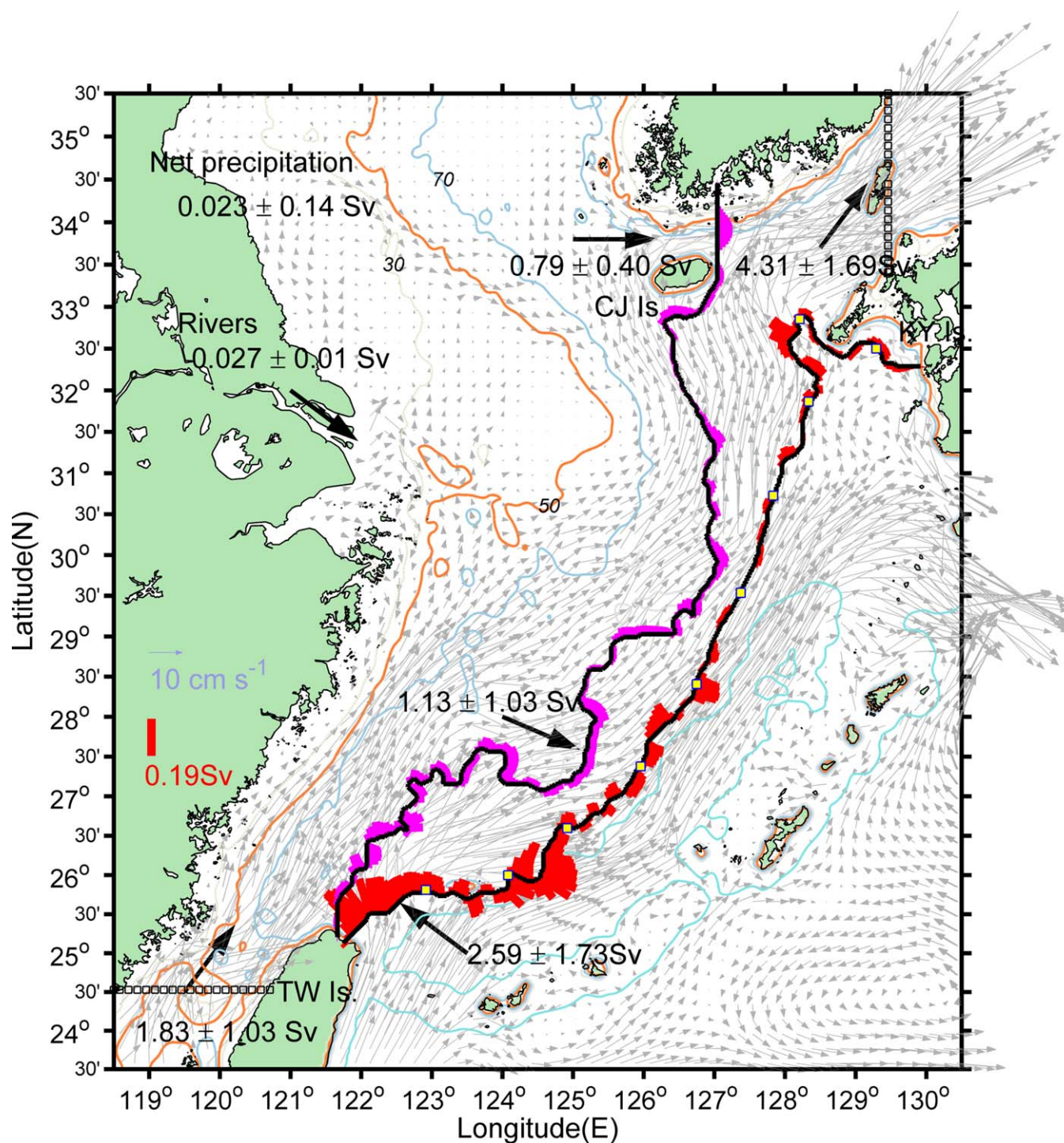


Figure 9. Annual mean water budget in the shelf of the East China Sea for 2006. The numbers show the net volume transport and their stand deviations through each transect (Unit: Sv) and the arrow indicates the direction of corresponding net volume transport. Exchanges through 100 m and 200 m isobaths (black curves) are also shown as bars in magenta and red, respectively. The yellow squares indicate the distance along the 200 m isobath from the Taiwan coastline with 150 km interval. The length of these bars indicates the magnitude of the annual mean transport at each model grid point across the isobath and the direction of these bars represents the onshore or offshore exchanges. The annual mean barotropic current is superimposed in light blue on the map and each arrow represents an average of neighboring 5×5 grid points.

Liu et al., 2014], this study found a significant northward shift of the onshore transport there from April to August (Figure 10). The modeled significant onshore intrusion near the northern end of the Okinawa Trough also differs from the previous simulations in that most onshore exchange occurred southwest of Kyushu

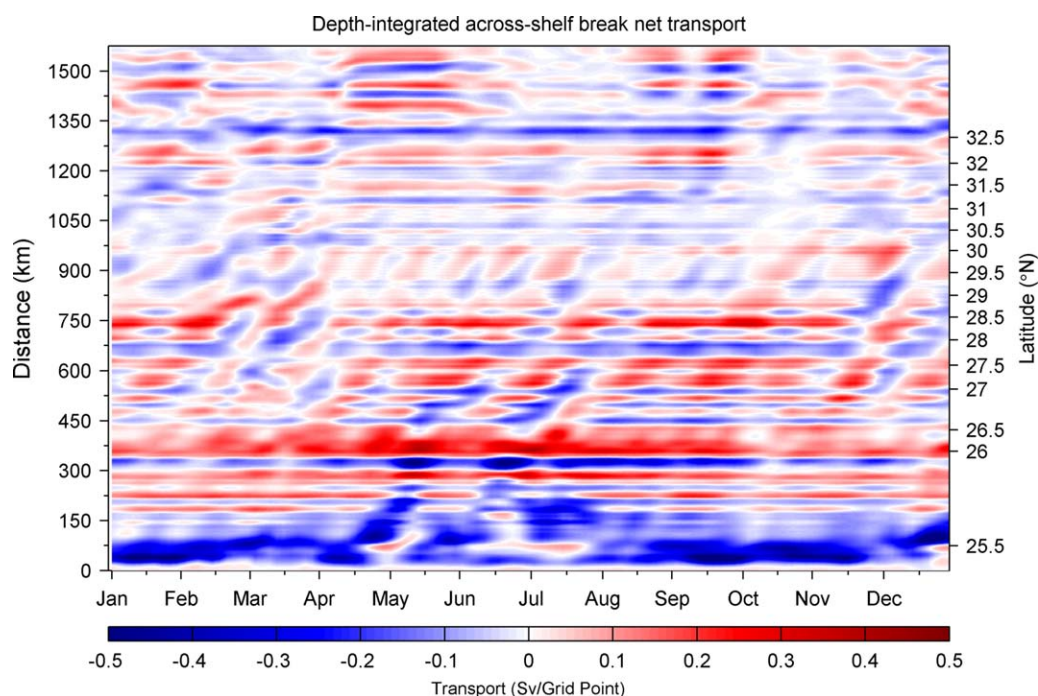


Figure 10. Variability of the vertically integrated transport across the shelf break for the hindcast.

where the shelf water was carried into the trough west of Kyushu. In the present model, the net exchange is almost steadily shoreward west of Kyushu (Figure 9). The disparity probably arises from the different choice of the fence and it is also due to the complicate circulation west of Kyushu where numerous eddies are generated as the Kuroshio separates. The exchange west of Kyushu can be demonstrated by trajectories of surface buoys. For example, buoy 5 and 6 entered into the northern ECS shelf (Figure 12b).

To analyze the causes responsible for the water exchange at the shelf break and further delivery of the mixture water in the ECS shelf, momentum equations are analyzed. An alongshore coordinate system is used

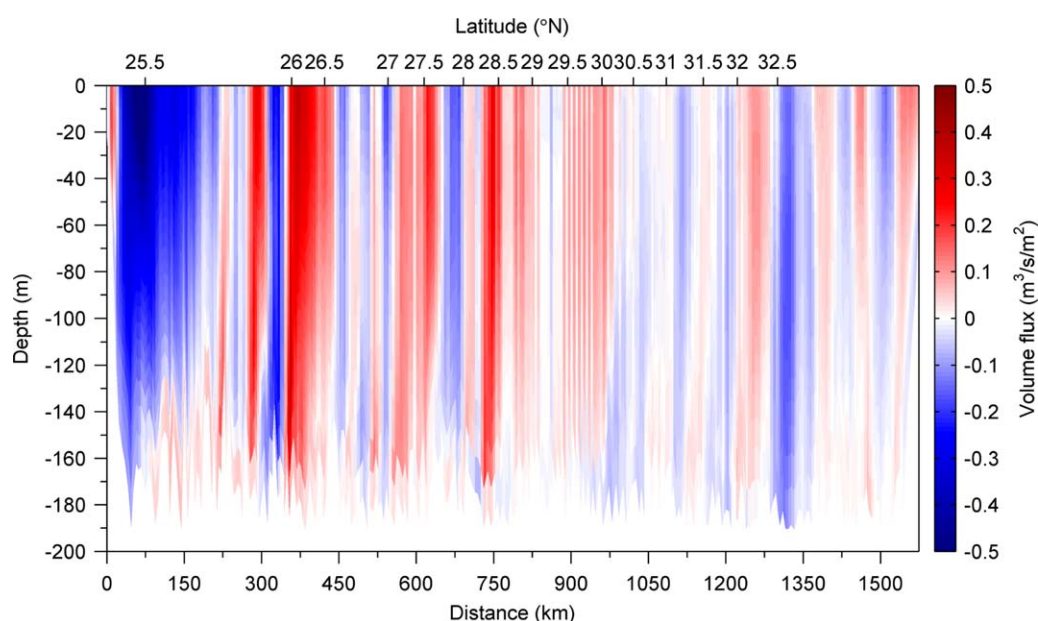


Figure 11. Annual mean volume flux across the shelf break indicated by the open squares in Figure 1.

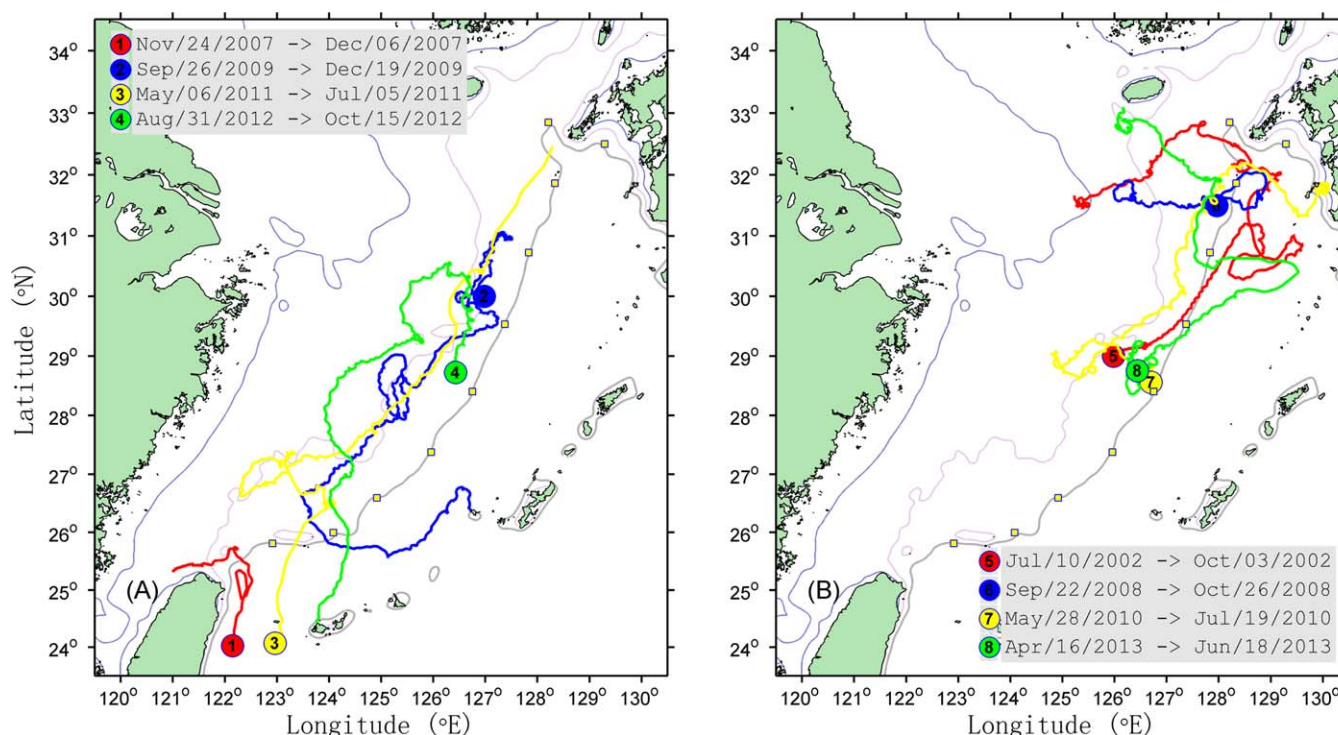


Figure 12. Trajectories of surface drifting buoys operated by Japan Meteorological Agency. The numbered circles represent the release location of the corresponding buoys.

with positive x pointing offshore and z upward. In the along-shelf direction, for example, the acceleration is due to terms on the right-hand side which are associated with Coriolis force ($-fu$, aka ua_cor), horizontal and vertical advection ($-\vec{u} \cdot \nabla v$, aka ua_hadv plus ua_vadv), pressure gradient ($-\frac{1}{\rho_0} \frac{\partial p}{\partial y}$, aka ua_prsgrd), horizontal viscosity, and vertical viscosity ($A_h \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + \frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right)$, aka ua_hvisc plus ua_vvisc), respectively (Figures 13a–13d). Among them, u represents cross-shelf velocity. While the analysis on the 3-dimensional equations reveals the distribution of individual terms, their relative contribution to the transport can be evaluated more clearly by the terms from the vertically integrated equations (Figures 13e–13g). It is noted that the vertical turbulent transport appears only as surface stress ($\frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right) |_{z=\eta}$, aka ua_sstr) and bottom stress ($\frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right) |_{z=-h}$, aka ua_bstr) in the depth-integrated equation.

As shown in Figures 11 and 13b, the barotropic signals are more discernable than the baroclinic counterparts which appear primarily northeast of Taiwan. The right-hand side terms suggest the cross-shelf momentum is mostly balanced by along-shelf pressure gradient force, which is the major contribution to exchanges over most of the shelf break (Figures 13a and 13b). The pressure gradient appears to be rather homogeneous in the vertical with slight variations discernable, differing from that presented by Guo *et al.* [2006]. The nonlinear effects indicated by advection terms are one order less than the pressure gradient force and essentially in the opposite direction of the latter (Figure 13c). Except for northeast of Taiwan, they are found to primarily induce the offshore transport that is noticeable in the southern shelf break but less significant in the middle and northern part of the shelf break. The viscous term owing to Ekman effects near the surface/seabed is significantly larger than that owing to internal shear so that the profile shows a sandwich-like structure in the vertical (Figure 13d). Their mean magnitude is less than the nonlinear terms and two orders smaller than the geostrophic terms. The surface Ekman stress, facilitating onshore transport, affects only upper layers with nearly uniform spatial distribution (Figure 13g), being analogous with Liu and Gan [2012]. The stress at the bottom differs from that at the surface, which is larger than the surface term and shows evident spatial variations, especially in the southern part of the shelf break. It essentially assists the export northeast of Taiwan and in the northern end of Okinawa Trough, and the import in the middle of the shelf break fence where bottom stress is negative.

Additional analysis based on the 100 m isobath gave rise to a similar result (not shown) and will not be discussed further. Overall, the general diagnosis at the ECS shelf break is consistent with the study presented

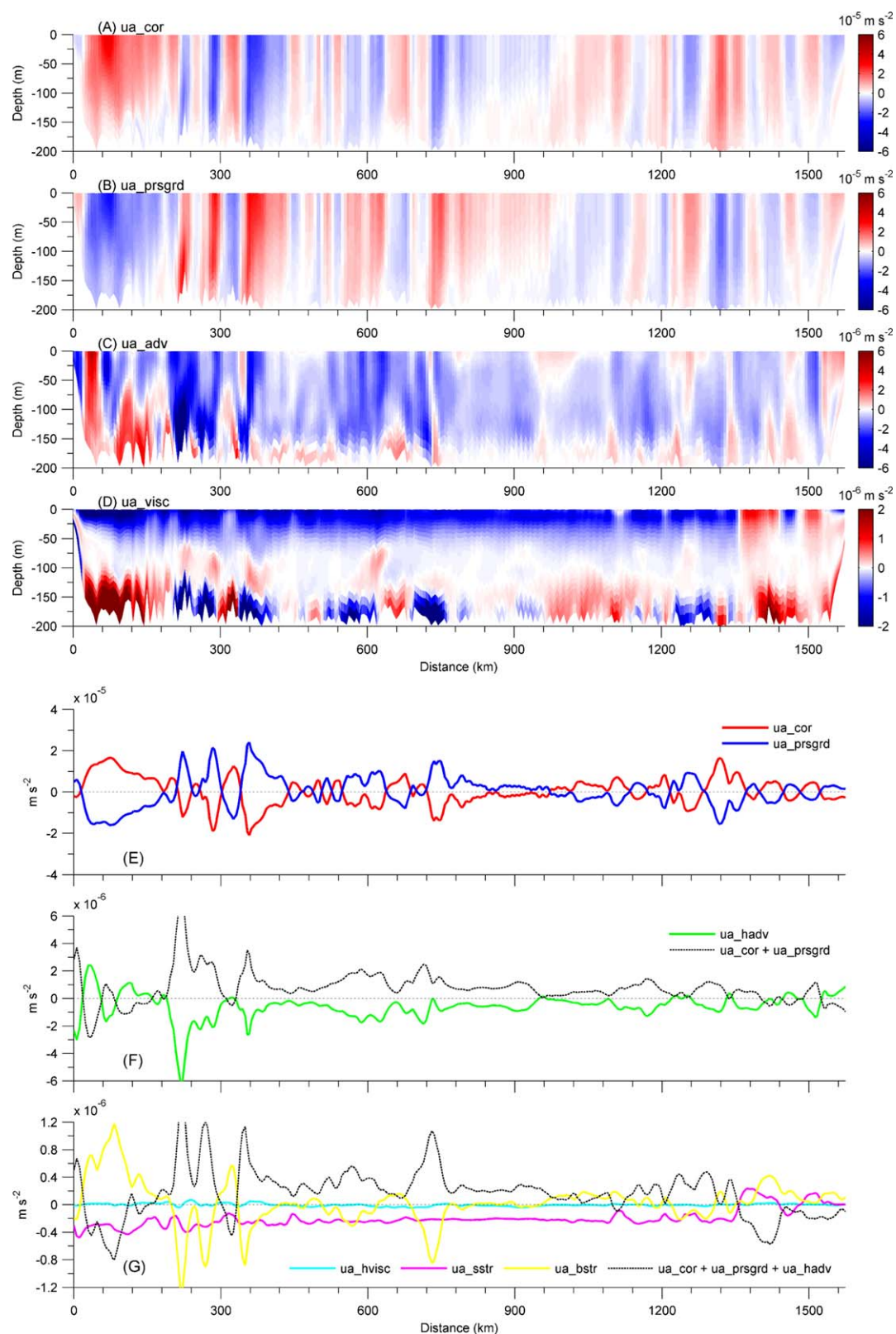


Figure 13. Annual mean of the right hand side terms of the momentum equations (top plot) and the depth-integrated equations (bottom plot) from the along-shelf direction at the shelf break, with northeastward being positive.

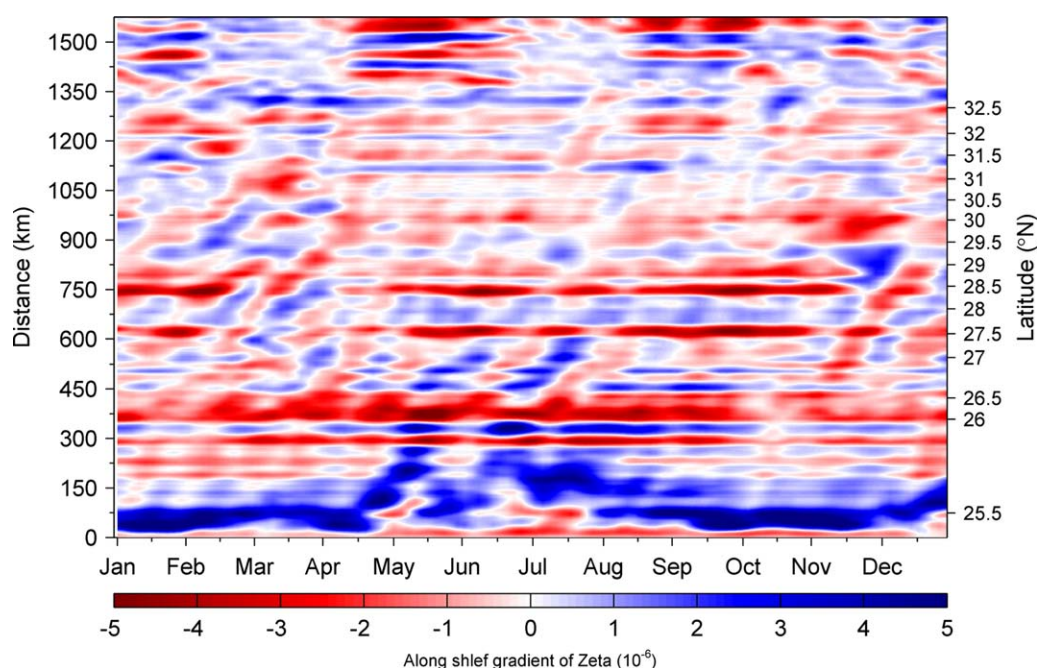


Figure 14. Distribution of the along-shelf-break gradient of the modelled sea level. The positive sea level gradient (blue) is generally responsible for onshore transport.

by Zhang *et al.* [2011] on the New England shelf break such that across-shelf momentum balance in the outer shelf is predominantly geostrophic. On the other hand, the Ekman drift is larger than that in the northwestern Black Sea shelf break as seen by Zhou *et al.* [2014], which focused on the cross-shelf break exchange in the warm seasons when wind forcing was relatively weak.

4.2. Variability of the Cross-Shelf Exchanges in 2006

The cross-shelf exchange is highly variable as shown in Figures 7c and 10. Further analysis illustrates the variability of the right-hand side terms. Since the pressure gradient is the primary factor that responsible for the onshore transport, which is aided by friction, it is worthwhile to understand their variability.

The model produces the sea level elevation, equals to the dynamic height associated with the barotropic pressure gradient force. The time series of the along-shelf-break gradient of the sea level elevation (Figure 14) resembles that of the vertically integrated transport through the shelf break (Figure 10), hence suggesting the importance of the pressure gradient on the cross-shelf exchange. On the shelf break fence from Taiwan northward to 200 km, the sea level rises fast to the maximum, which drives a considerable amount of Kuroshio water shoreward. From May to July, both of these two figures signal a northward shift of the shoreward intrusion. The outgoing locations also shifted northward similarly. In the northern Okinawa Trough, the two figures both suggest a narrow band of onshore movement that is consistent with the float trajectory shown in Figure 12b and the model study presented by Guo *et al.* [2006, their Figures 4–6].

The mean surface Ekman-layer depth at the shelf break ranged from 21 m in summer to 90 m in winter, yielding an annual mean of 58 m for 2006 based on the formula presented by Stewart [2008, p. 142].

$$D_E = 7.12 U_{10} \sqrt{\sin \varphi}$$

Here U_{10} is the wind speed at 10 m above the sea surface, and φ is latitude. It is noted that the Ekman depth estimation is also similar to those shown in Figures 13d and 13h.

The spatially averaged wind vectors over the ECS shelf were featured by northeasterly monsoon in the winter, exceeding the southerly monsoon in the summer on the average (Figure 15a), hence the annual mean surface stress resembles that in winter. The winter monsoon enhanced the onshore exchange in upper layers (0–50 m), about 1.26 Sv larger than that offshore on the annual average (Figures 15b and 16d). In summer, the southerly monsoon contributes mostly to the export except for typhoon periods. The

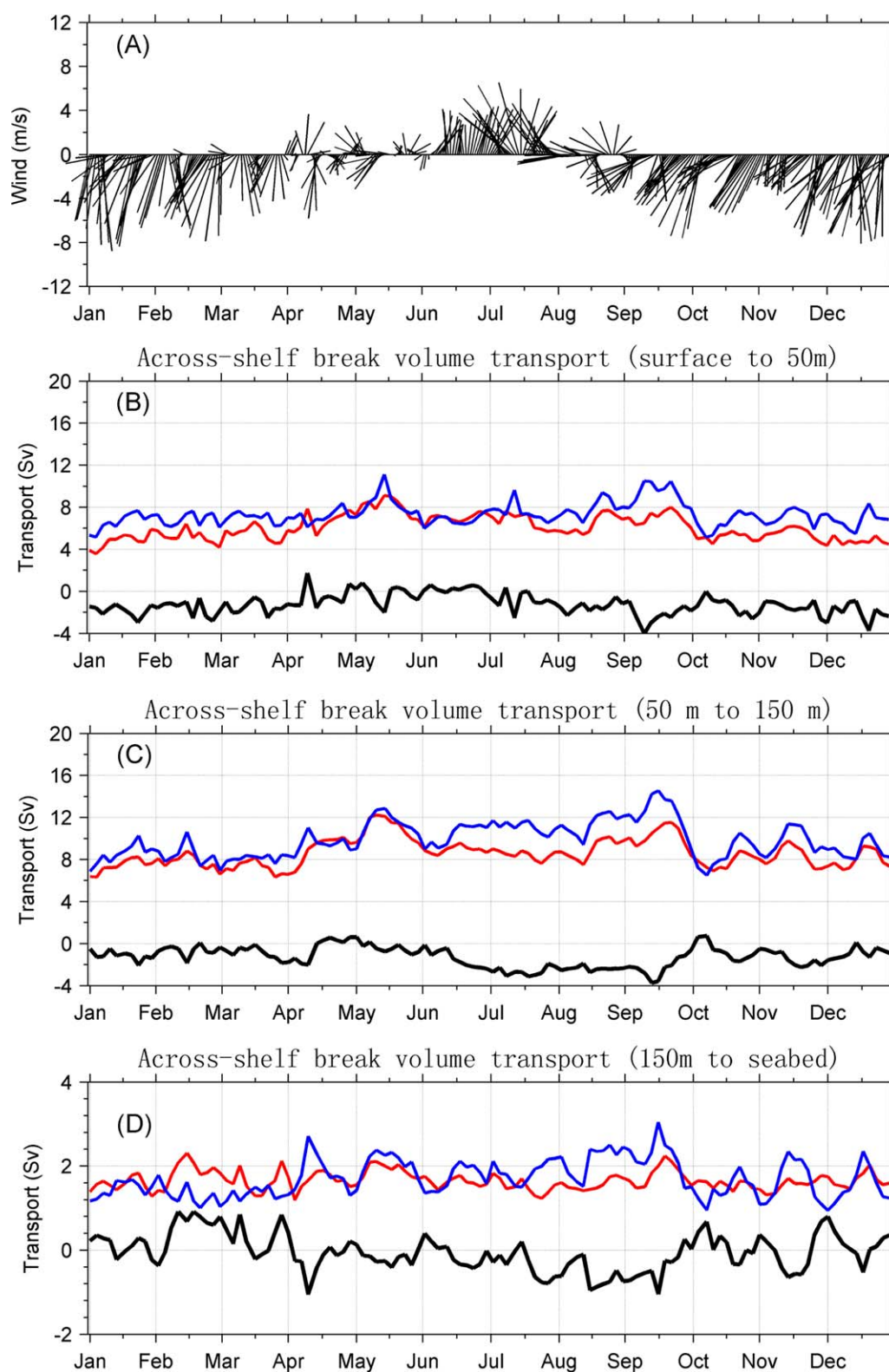


Figure 15. Wind vectors and the cross-shelf break transport. Daily wind (a) is averaged on the shelf of the East China Sea. Transports, integrated within upper layers (b: 0–50 m), middle layers (c: 50 m–150 m), and lower layers (d: 150 m–seabed) respectively, are shown as offshore (red), onshore (blue, absolute value), and net (black, positive for offshore and negative for onshore).

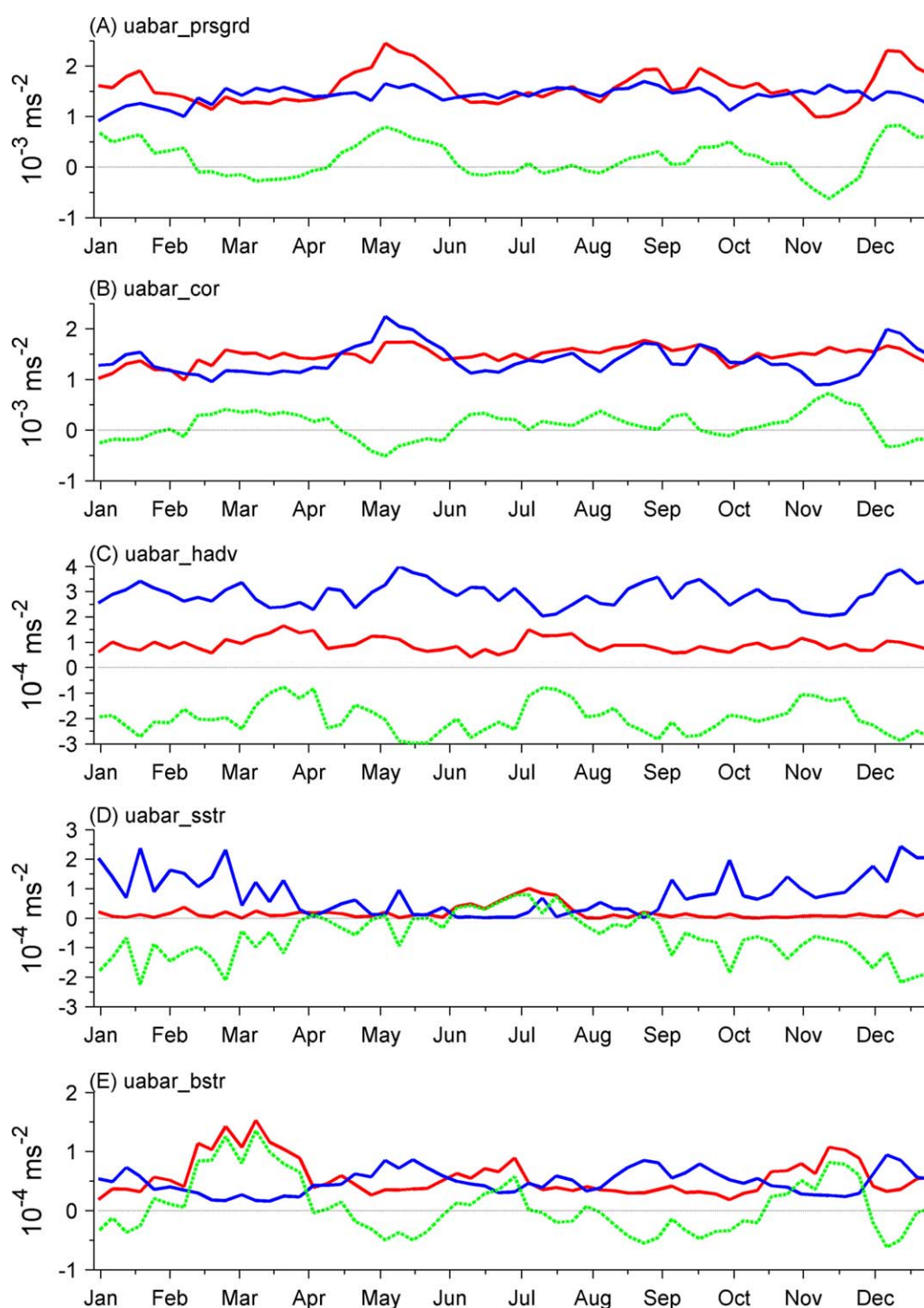


Figure 16. Time series of the right hand side terms of the momentum equations from the along-shelf direction, which were integrated along the shelf break from the depth-integrated equations. Northeastward is positive in red lines and southwestward are in blue lines (absolute value), with their sum in green lines.

correlation coefficient between the wind and the upper layer transport is 0.58 with almost no phase lags, suggesting the Ekman effects to some extent are responsible for variability in the upper layer exchanges despite that on the annual mean basis the primary balance is geostrophic. Reduction of the integration depth has little impacts on the correlation, but increase of the integration depth makes the correlation weaker. The correlation coefficient reduces to only 0.39 if the transport is integrated over the entire water

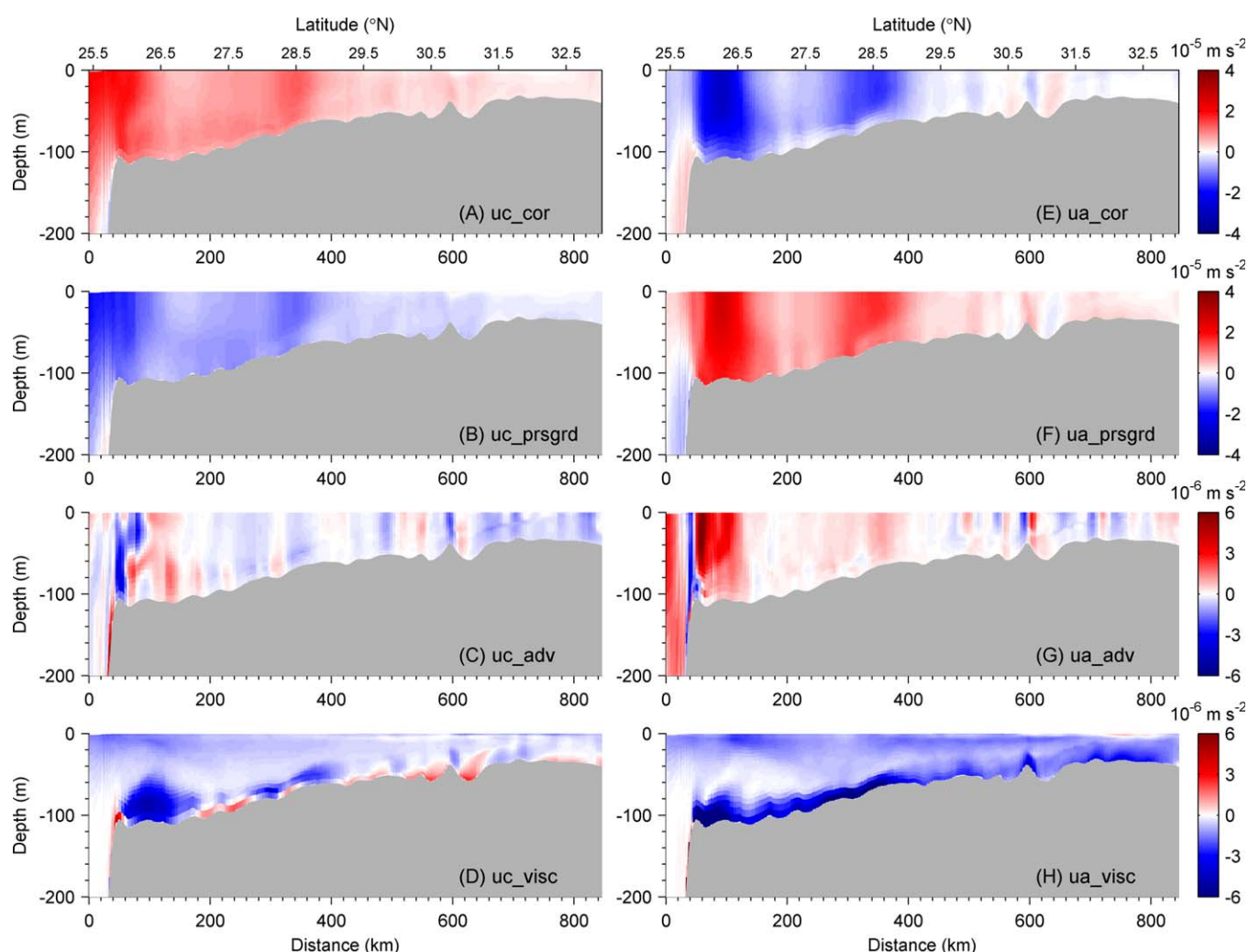


Figure 17. Annual mean distribution of the right hand side terms of the momentum equations (m s^{-2}) along the transect TY as shown in Figure 5. Left/right plots indicate individual terms from the cross-shelf/along-shelf equation, respectively.

column. During the transition period of April–May and the weak southerly wind of June–mid-July, the cross-shelf exchanges approached more or less a net balance. It is also noted that the relax of tropical storm Bilis enhanced the inward transport around 15 July.

In the interior water column, the annual mean onshore transport exceeded the offshore counterpart by 1.28 Sv and the difference occurred mostly in the summer and early fall (Figure 15c). In comparison with the upper layers where the local wind stress took effects, the cross-shelf exchange from the interior water column was mostly driven by the sea level elevations in responses to the large-scale wind stress curl field.

In lower layers from 150 m to the sea floor, inward and outward exchanges almost balance each other for large part of the year (Figure 15d). The annual mean net exchange was onshore at only 0.06 Sv. Seasonally, the net exchange was onshore and mostly occurred in the spring and summer, while it was offshore in the winter. The temporal variation of the low layers shoreward exchange was to some extent consistent with the variability of the negative bottom stress integrated along the shelf break with the correlation coefficient being at 0.47 (Figure 16e).

Overall, the cross-shelf exchange is essentially govern by geostrophic balance but is also assisted by the nonlinear effects and Ekman stress (Figure 16). Particularly in terms of net cross-shelf contribution, nonlinear effects affect persistently on the offshore transport throughout the year (Figure 16c). The surface stress contributes to onshore transport mostly from September to mid-May because of strong northerly wind which

prevails in winter but episodically occurs in spring and early fall (Figure 16d). The bottom stress facilitates the onshore transport episodically, for instance during the periods of December–January, May, August through September, while aids the offshore transport around March (Figure 16e).

4.3. TWC in the Inner Shelf

In spite of large amount of water in and out of the shelf break, only about 10% the swapped water is left on the shelf to be mixed with the TWS water, which in turn evolves to the well-known shelf flow—the TWC. The northward excursion of the TWC has been well documented with field measurements [Guan, 1984; Weng and Wang, 1984] and numerical simulations [Yang *et al.*, 2012]. The survey data show that the TWC carries saline water to the northern ECS with their nearshore branch approaching north of the CR mouth as shown in Figure 4a, which was rare during normal years. This stronger-than-normal northward penetration was also reproduced by the model as shown in Figures 4b–4d, 5, and 9.

From the annual mean perspective, analysis revealed that geostrophic terms set up the leading-order momentum balance at most of the transect TY in 2006 (Figure 17), similar to the case at the shelf break. The Coriolis force associated with the northward TWC was balanced by the cross-shelf pressure gradient force. The Ekman stress both at the surface and at the bottom was against the TWC and favored shoreward cross-shelf transport.

However, the pressure gradient force along the shelf break associated with the large-scale general circulation may not be able to drive the cross-shelf exchange to the north of the CR mouth in normal years, because the pressure gradient force does not penetrate that far onto the shelf. Figure 5 suggests that the northward excursion of the TWC in 2006 was not only earlier and stronger but also persisted longer than normal years as indicated by the 34 and 34.5 isohalines. The enhanced TWC can be attributed to the fact that both the cross-shelf exchange through the shelf break and the flow through the TWS were more intense in 2006 than normal years (Table 2 versus Table 1). Figure 9 could explain how the midshelf interacted with the outer shelf by showing exchanges across 100 m isobath in 2006. The exchange magnitude was considerably larger in both the southern and the northern shelf. The TWC, as a mixture between the TWS water and the intruded Kuroshio water, might have formed everywhere south of 29°N, where both onshore and offshore exchanges were strong.

It is noted that the stronger intrusion of the TWC than normal years could be affected by the difference in the forcing driving the two runs. However, this artifact was limited in our study. First, in the sense of climatological mean, both wind fields are in good agreement either in winter when monsoon is strong or in summer when monsoon is rather weak (not shown). Hence, the sea level elevation along the shelf break driven by the large-scale wind fields and Ekman effects driven by local winds should be similar. Second, the difference of the heat flux between the COADS and the climatological ECMWF appears relatively large in the ECS, but the correction of the heat flux with respect to the SST should constrain their difference because the SST product both from COADS and ECMWF are adapted from observations. Further, the comparison of the cross-shelf volume transport between the climatological run (−1.17 Sv) and the hindcast run (−2.59 Sv) is overwhelmingly significant.

In the northern ECS shelf, calculation at a transect along 31°N off the CR mouth suggests that the annual mean northward volume transport associated with the inner branch of the TWC reaches 0.01 Sv, which was almost identical to that of southward coastal current ZMCC and responsible for the high-salinity and low-temperature water within the submerged river valley as shown in Figure 3. Presumably, the TWC could also bring high-ratio-phosphorus and low-oxygen-concentration water into the northern shelf [Liu *et al.*, 1992; Chen, 1996], stimulated the local biological production and intensified the reduction of oxygen at depth, which could be partially responsible for the hypoxia in the northern ECS as suggested by Li *et al.* [2002] and Wei *et al.* [2007].

Besides the exchange occurring at the shelf break, the flow through the TWS is also important to the density change in the ECS shelf. Though its magnitude through the TWS is nearly one order less than that through the shelf break, the net volumetric transport, heat, and salt transport entering the ECS via the TWS is comparable to that from the latter. As suggested in Figures 6 and 7, the seasonal variability of the exchange at the shelf break was weak. Comparatively, the exchange at the TWS has relatively strong seasonality, with the northward flow prevailing in summer but episodically occurring in winter. Therefore, the TWC would show temporal variations of the composite of the cross-shelf exchange and the flows through the TWS depending

on their relative magnitudes and time lines. It could be inferred that the TWC should affect the circulation and the biogeochemical budget on the ECS shelf with multitime scales, which requires further attention.

5. Conclusions

This study quantifies the volume transport across the boundaries of the ECS and analyzes the associated dynamic balance from a climatological experiment and the 2006 hindcast run using a 3-D high-resolution ocean circulation model. Both observations and simulations find that high-saline water was delivered to the north of the CR mouth in 2006 (near 32°N), which is farther north than in normal years reported as a result of stronger cross-shelf exchanges and flow through the TWS than normal years. The annual mean rate of the net across-shelf break volumetric exchange is 2.59 Sv in the 2006 hindcast, up 120% on the climatological experiment, and is 40% larger than the transport from the TWS, which is up 20% on the climatological mean. The enhanced TWC intrusion could have significant effects on the ecosystems off the CRE by regulating the distribution of nutrients and affecting stratification, and is likely responsible for the northward spread of the hypoxic zone.

A few new places on the shelf break are identified as locations of persistent onshore or offshore exchanges throughout the year. Areas south of 26°N and south of 28°N contribute significant onshore transport, in addition to the previously well-documented onshore intrusions northeast of Taiwan and in the northern end of the Okinawa Trough. On the other hand, shelf break from 26.0°–26.5°N, around 27.5°N and 28.5°N are resolved as major outgoing places. Comparing with the stable locations for the exchanges above, the onshore transport northeast of Taiwan shows large seasonality, highlighted by the northward shift from April to August. Further field and modeling activities at these places would be essential to understand the interannual variability of these key channels of cross-shelf exchanges and their implications for the ecosystem of the ECS.

The cross-shelf exchange depends heavily on the along-shelf geostrophic balance for the total cross-shelf transport. The barotropic pressure gradient force is stronger than the baroclinic portion in the cross-slope transport, especially near the southern end of the shelf break. The sandwich-like Ekman stress profile results from the monsoon wind at the surface and the friction at the bottom. The annual mean net onshore exchange in upper layers is mostly affected by northeasterly wind, which is strong from fall to late spring. By contrast, the net transport induced by advection effects is seaward.

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